SIEMENS

AK 1703 ACP

Functional Description

MC2-011-2.01

Hint

Please observe Notes and Warnings for your own safety in the Preface.

Disclaimer of Liability
Although we have carefully checked the contents of this publication for conformity with the hardware and software described, we cannot guarantee complete conformity since errors cannot be excluded. The information provided in this manual is checked at regular intervals and any corrections that might become necessary are included in the next releases. Any suggestions for improvement are welcome.

Subject to change without prior notice.
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Preface

This document is applicable to the following product:

- AK 1703 ACP

Purpose of this manual

This manual describes the function and the manner of working of the AK 1703 ACP systems and gives instructions for installation. It is intended for persons given in the next section.

For the various activities the manual provides diversely detailed information and overviews.

- In order to get a general view of the product and its applications
  - chapter "Application Overview"
- More detailed information on the functionality of the system and on how the individual parts work together are given in
  - chapter "Function Packages", sections "Telecontrol", "Automation", and "System Services"

  This information may be of help whenever, caused by poor understanding, questions arise during the engineering work
- Instructions for the installation of the device
  - chapter "Mechanical Design and Installation"
- Technical Specifications
  - chapter "Technical Specification of the System"
  - section "Technical Specification" of the respective system element manual (see Literature)
- Detailed information on the engineering of the individual system elements can be found in the system element manuals (see Literature)

Target Group

This manual is intended for persons who are entrusted with installation, design, parameter setup and programming of AK 1703 ACP systems.

Notes on Safety

This manual does not constitute a complete catalog of all safety measures required for operating the equipment (module, device) in question because special operating conditions might require additional measures. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger.

Danger

means that death, serious bodily injury or considerable property damage **will** occur, if the appropriate precautionary measures are not carried out.

Warning

means that death, serious bodily injury or considerable property damage **can** occur, if the appropriate precautionary measures are not carried out.

Caution

means that minor bodily injury or property damage could occur, if the appropriate precautionary measures are not carried out.

Hint

is important information about the product, the handling of the product or the respective part of the documentation, to which special attention is to be given.

Qualified Personnel

Commissioning and operation of the equipment (module, device) described in this manual must be performed by qualified personnel only. As used in the safety notes contained in this manual, qualified personnel are those persons who are authorized to commission, release, ground, and tag devices, systems, and electrical circuits in accordance with safety standards.

Use as Prescribed

The equipment (device, module) must not be used for any other purposes than those described in the Catalog and the Technical Description. If it is used together with third-party devices and components, these must be recommended or approved by Siemens.

Correct and safe operation of the product requires adequate transportation, storage, installation, and mounting as well as appropriate use and maintenance.

During operation of electrical equipment, it is unavoidable that certain parts of this equipment will carry dangerous voltages. Severe injury or damage to property can occur if the appropriate measures are not taken:

- Before making any connections at all, ground the equipment at the PE terminal.
- Hazardous voltages can be present on all switching components connected to the power supply.
- Even after the supply voltage has been disconnected, hazardous voltages can still be present in the equipment (capacitor storage).
- Equipment with current transformer circuits must not be operated while open.
- The limit values indicated in the manual or the operating instructions must not be exceeded; that also applies to testing and commissioning.

Typografic and Sign Conventions

- Manuals to be referred to are represented in italics, such as e.g. *ACP 1703 Common Functions System and Basic System Elements*.
- For easy reading, certain designations and names are presented *in this font*
- Symbolic names are presented *in this font*

Table of Contents

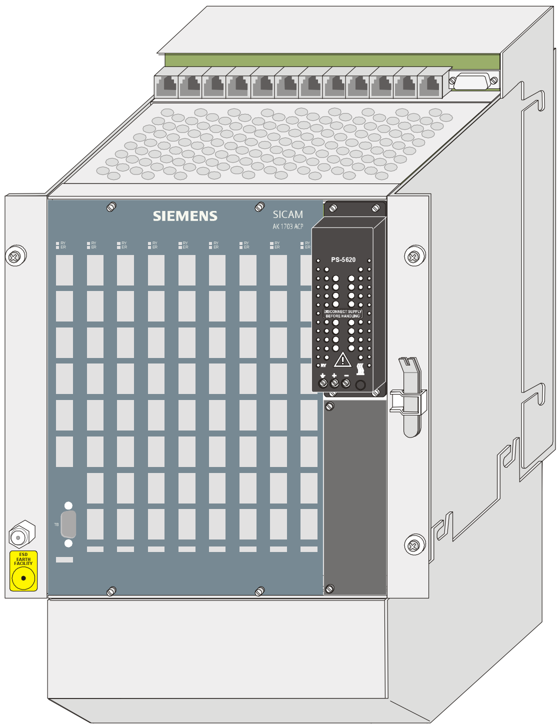
1. Introduction

Contents

1.1. Longevity through Continuity and Innovation

Following the principle of our product development, **AK 1703 ACP** has high functionality and flexibility, through the implementation of innovative and reliable technologies, on the stable basis of a reliable product platform.

For this, the system concept ACP (Automation, Control and Protection) creates the technological preconditions. Balanced functionality permits the flexible combination of automation, telecontrol and communication tasks. Complemented with the scalable performance and various redundancy configurations, an optimal adaptation to the respective requirements of the process is achieved.



AK 1703 ACP is thus perfectly suitable for automation with integrated telecontrol technology as:

- Telecontrol substation or central device
- Automation unit with autonomous functional groups
- Data node, station control device, front-end or gateway
- With local or remote peripherals
- For rear panel installation or 19 inch assembly

1.2. AK 1703 ACP - the forward-looking product

- **Branch-neutral product, therefore versatile fields of application and high product stability**

  Fields of application:

  - Hydroelectric power stations (turbine governor, process control)
  - Electrical energy distribution and transmission
  - Oil/Gas pipelines
  - Tunnels

- **Versatile communication**
  - up to 66 serial interfaces according to IEC 60870-5-101/103
  - LAN/WAN communication according to IEC 60870-5-104
  - various third-party protocols possible

- **Easy engineering**
  - TOOLBOX II
  - Object-orientation
  - Creation of open- and closed-loop control application programs according to IEC 61131-3
  - all engineering tasks can also be carried out remotely

- **Plug & play for spare parts**
  - Storage of parameters and firmware on a Flash Card
  - spare part exchange does not require additional loading with TOOLBOX II

- **Open system architecture**
  - Modular, open and technology-independent system structure
  - System-consistent further development and therefore an innovative and future-proof product

- **Scalable redundancy**
  - Component redundancy
  - Doubling of processing/communication elements

- **The intelligent terminal - TM 1703**
  - Direct connection of actuators and sensors with wire cross-sections up to 2.5mm²
  - Can be located remotely up to 200m
  - Binary input/output also for 110/220VDC
  - Assembly on 35mm DIN rail

1.2.1. Versatile communication capability

With AK 1703 ACP, a variety of media can be utilized for local and remote communication. (wire connections, FO, radio, dial-up traffic, GSM, GPRS, WAN, LAN, field bus etc.)

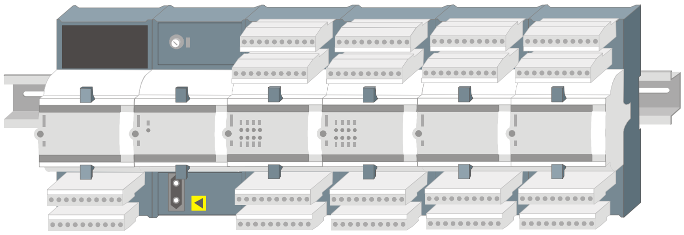
Through the simple installation of serial interface modules, in total up to 66 communication interfaces are possible in one AK 1703 ACP, whereby a different individual protocol can be used for each interface.

For standard communication, protocols according to IEC 60870-5-101/103/104 are implemented. The consistent implementation of the IEC 60870-5-101/103/104 standard guarantees a uniform addressing from the source through to the sink.

Besides the protocols according to IEC 60870-5-101/103/104, there are also a variety of third-party protocols available (DNP3.0, Modbus etc.). Through this, the seamless integration into existing automation networks is enabled, whereby a long-term safeguarding of already effected investments is ensured.

1.2.2. Simple process interfacing

In addition to the central acquisition and output of process signals within an AK 1703 ACP mounting rack, it is possible to use TM 1703 peripheral elements.



An essential feature of the TM 1703 peripheral elements is the efficient and simple interfacing possibility of the process signals. This takes place on so-called I/O modules, which are distinguished through a robust casing, a secure contact as well as solid electronics. The I/O modules are lined up in rows. The contact takes place during the process of latching together, without any further manipulation. Thereby each module remains individually exchangeable.

A clearly arranged connection front with LEDs for the status display ensures clarity locally. The structure of the terminals enables a direct sensor/actuator wiring without using intermediate terminals with wire cross-sections up to 2.5 mm^2. Modules for binary inputs and outputs up to 220 VDC open further saving potentials at the interface level.

Depending on the requirements, the I/O modules can be fitted with either an electrical bus or an optical bus, through which the peripheral signals can be acquired as close as possible to the point of origin. In this way a broad cabling can be reduced to a minimum.

1.2.3. Easy engineering

An essential aspect in the overall economical consideration are the costs that occur for the creation, maintenance and service. For this, the reliable **TOOLBOX II** is used.

- **Object-orientation**

  The object-orientation makes it possible to also utilize the same characteristics of same-type primary-technology units and operational equipment (e.g. disconnectors, circuit breakers, feeders etc.) for the configuration. The close coupling with the design tool ensures the consistent, uniform documentation of the entire plant through to circuit diagram.

  Through this, considerable rationalization results with engineering.

- **Open-loop and closed-loop control according to IEC 61131-3**

  Open- and closed-loop control application programs are created by means of CAEx *plus* according to IEC 61131-3, a standard that is generally accepted and recognized in the market. As a result, the training periods are reduced considerably.

- **All engineering tasks can also be carried out remotely**

  All engineering tasks, from the system diagnostic through to the online test, can also be performed remotely with the TOOLBOX II. For this, a separate communication link between TOOLBOX II and AK 1703 ACP is not necessary: every available communication interface can be used. Using further automation units of the ACP 1703 product family, the TOOLBOX II can be remotely positioned over an arbitrary number of hierarchies.

The access to the engineering data is fundamentally protected by a password.

Plug & play for spare parts

Flash Card for data storage

All data of an automation unit - such as firmware and parameters - are stored non-volatile centrally on an exchangeable Flash Card. With a restart of the automation unit, and also with a restart of individual modules, all necessary data are automatically transferred from the Flash Card to all CPUs and modules.

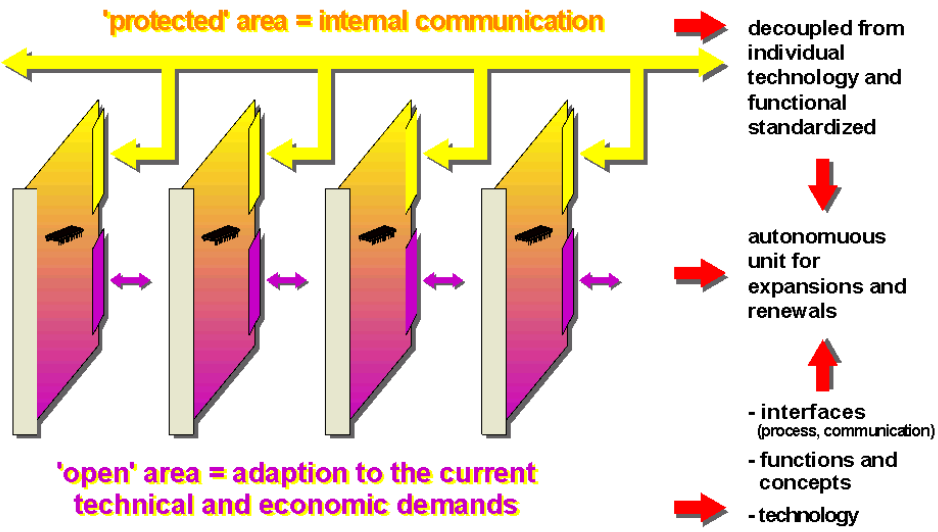
Consequently, with the exchange of modules, new loading is no longer required, since new modules obtain all data from the storage card. With the replacement of spare parts, Plug & Play becomes a reality: one needs no special tool for this, even loading is no longer necessary.

Thereby, work during a service operation is reduced to a minimum.

1.2.4. Open system architecture

The basis for this automation concept is a modular, open and consequently technology-independent system architecture for processing, communication and peripherals (multi-processor system, firmware).

Standardized interfaces between the individual elements again permit, even with further developments, the latest state of technology to be implemented, without having to modify the existing elements. In this way, a longevity of the product and consequently investment security and continuity can be ensured.



Every board and every module on which a firmware can run, forms, together with the function-determining firmware, one system element.

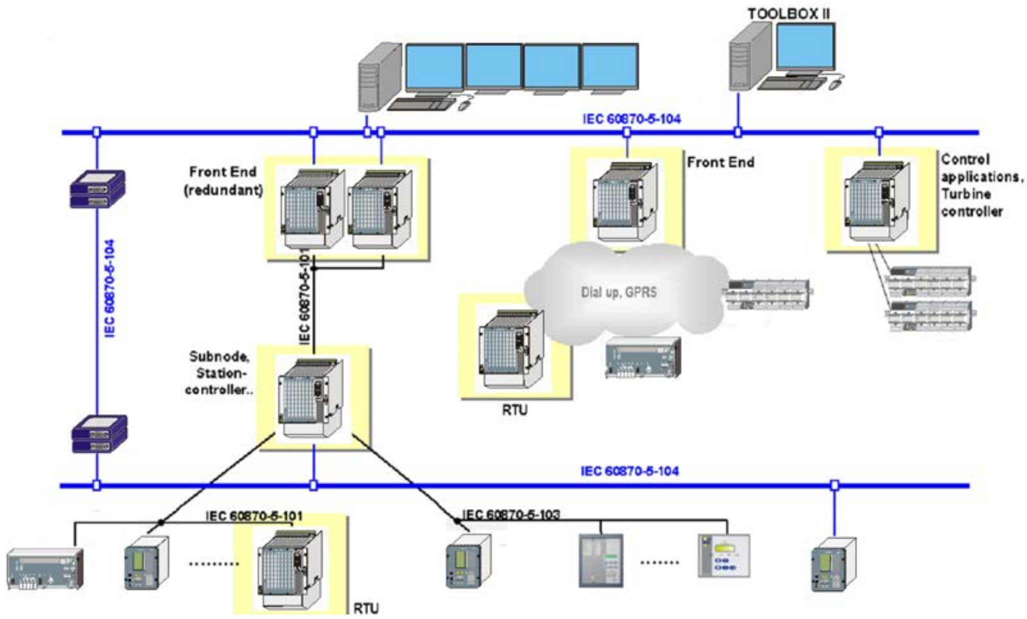
The adaptation to the specific requirements of the application is achieved through the individual configuration and through the loading of standard firmware and parameters. Within their defined limits, the parameters thereby not only influence the behavior of the firmware functions, but also that of the hardware functions. With that, for all module types, all mechanical parameter settings are omitted, such as e.g. the changing of jumpers or loads and thus enables not only the online change, but also a consistent documentation of the set parameters by the TOOLBOX II as well as a simplified storage.

2. Application Overview

Contents

2.1. Introduction

Due to the modular architecture, AK 1703 ACP can be used in a variety of ways:



- Front End, Gateway
- Process control applications, automation applications, turbine governor
- Station control device
- Sub-node
- Telecontrol substation
- Redundant configurations

In principle, for this all necessary functionalities are available. The actual application is defined simply through the corresponding configuration and parameterization.

2.2. Front End

Due to the large number of interfaces (up to 66) and the variety of protocols available, AK 1703 ACP is perfectly suitable for the use as front end for a process control system.

All telecontrol substations – regardless of which manufacturer and over which protocol – are connected to AK 1703 ACP. In the front end, the signal processing and adaptation takes place for the respective control system. From the perspective of the control system, there is no difference which protocol and which system behavior the substation actually has.

2.3. Process Control Applications, Automation Applications, Turbine Governor

Open- and closed-loop control application programs are created by means of CAEx *plus* according to IEC 61131-3, a standard that is generally accepted and recognized in the market.

In AK 1703 ACP, at every slot a system element can be fitted with *open-/closed-loop control function*. Through this and due to the modularity, AK 1703 ACP is suitable for many applications: from smaller automation applications through turbine governor up to complex process control applications. Naturally, all applications can also be combined.

2.4. Station Control Device, Sub-Node

The functionality of a station control device can be simply regarded as a combination of the functionality of a front end (interfacing of diverse bay devices, protective devices, processing of the data for the power system control) and the functionality of process control applications (open- and closed loop control application programs), and is therefore perfectly suited for this application. In addition, further telecontrol peripherals could also be installed in the station control device, through which telecontrol station and station control device could be united in one device.

2.5. Telecontrol Substation

For telecontrol applications there is a modular, versatile periphery available for the process data interfacing.

Especially due to the possibility of being able to remotely locate TM 1703 peripherals, AK 1703 ACP supports peripheral elements installed centrally and decentralized. Flexible communication functions also permit redundant communication and communication over stand-by transmission lines.

Naturally, arbitrary open- and closed-loop control application programs can be realized in AK 1703 ACP with CAEx *plus*, through which, at the same time and to the same degree, AK 1703 ACP can become a remote terminal unit and an automation unit in one.

2.6. Redundant Configurations

In order to increase the availability of a plant or a plant section, one can design certain system parts redundant, in other words double.

In AK 1703 ACP, redundancy is available in various forms:

- redundant communication routes
  Communication connections normally have by far the highest failure rate
- redundant processing elements
- redundant automation units
- redundant total systems

With redundancy, one part, the *active* one, is utilized for the operation, while the other part is *Hot-Standby*.

With redundant communication routes, both communication interfaces can be used simultaneously for the purpose of load distribution. If one communication link fails, then all data are transmitted over the communication link still available.

With simple redundancy applications, the processing system elements are designed redundant, whereby one is *active*, the other is *Hot-Standby*. The *Hot-Standby* processing system element is continually calibrated, through which, with a failure of the *active* element, a switchover to the *Hot-Standby* can take place without operational interruption. In this way, for example redundant turbine governors can be realized simply and cheaply. Up to 5 redundant processing elements can be installed in one AK 1703 ACP. Since the power supply can also be designed redundant, an availability is achieved close to that with two separate automation units.

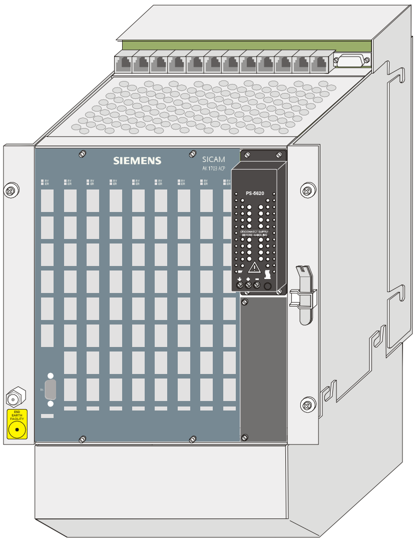
If the automation units are designed redundant, then a continuous calibration of the open- and closed-loop control application programs can also be performed.

For redundant front ends, the switchover can take place separately for each communication interface. This increases the availability, especially with redundant communication routes to the remote terminal unit.

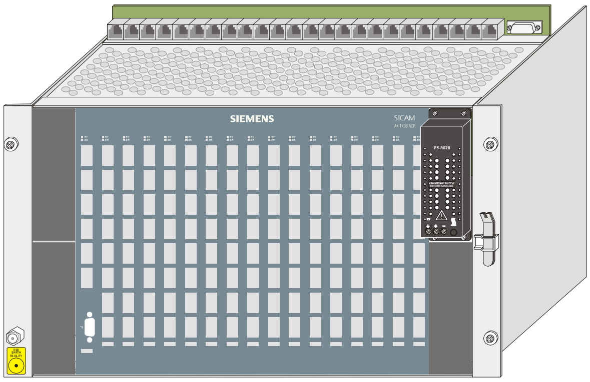
3. System Overview

Contents

3.1. Mechanics



9 Slots



17 Slots

Pictured above are two types of basic mounting racks:

	Designation	
CM-2832	AK 1703 ACP Mounting Rack with 9 Slots	
CM-2835	AK 1703 ACP Mounting Rack with 17 Slots	

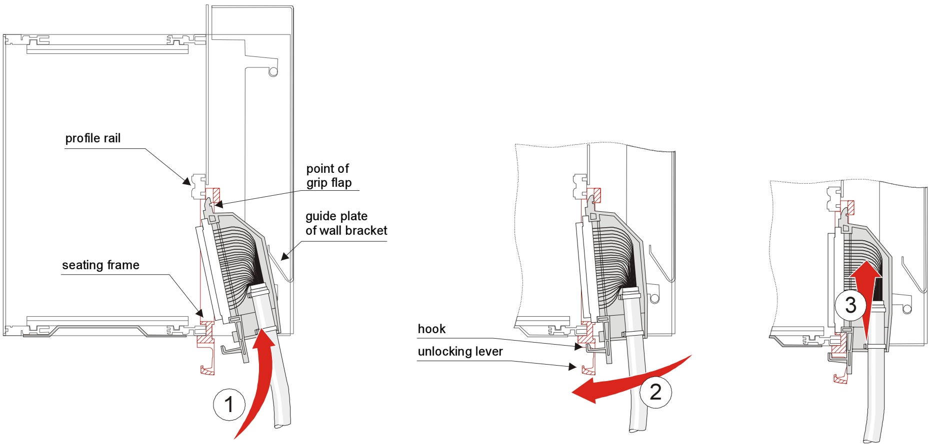
Not pictured here is the expansion mounting rack for up to 16 peripheral elements outside the basic mounting rack:

	Designation	
CM-2833	AK 1703 ACP Expansion Mounting Rack	optional

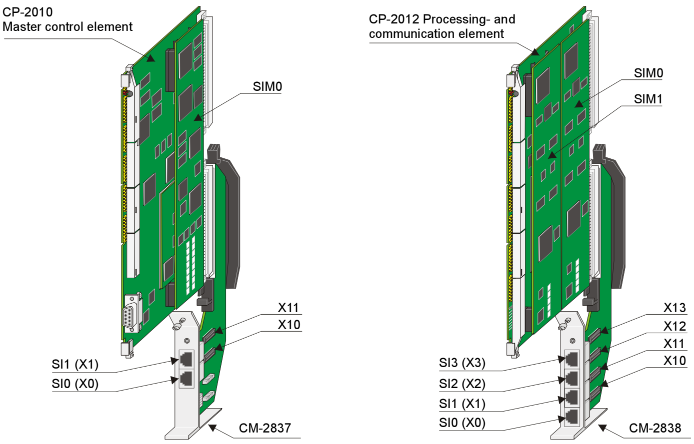
With the mechanics, value has been placed on flexibility and easy handling. Consequently, the mounting rack is available for **rear panel installation** or for **19" (swing) frame installation**, refer to Mechanical Design.

Almost all necessary external connectors (e.g. communication, peripherals, external periphery bus) can be connected with the help of standard cables or prefabricated cables without any additional tools.

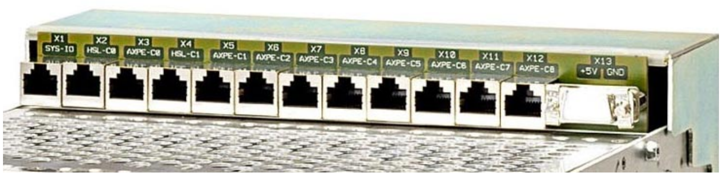
Connection Technique for Peripheral Signals



RJ45 Connection Technique for Communication



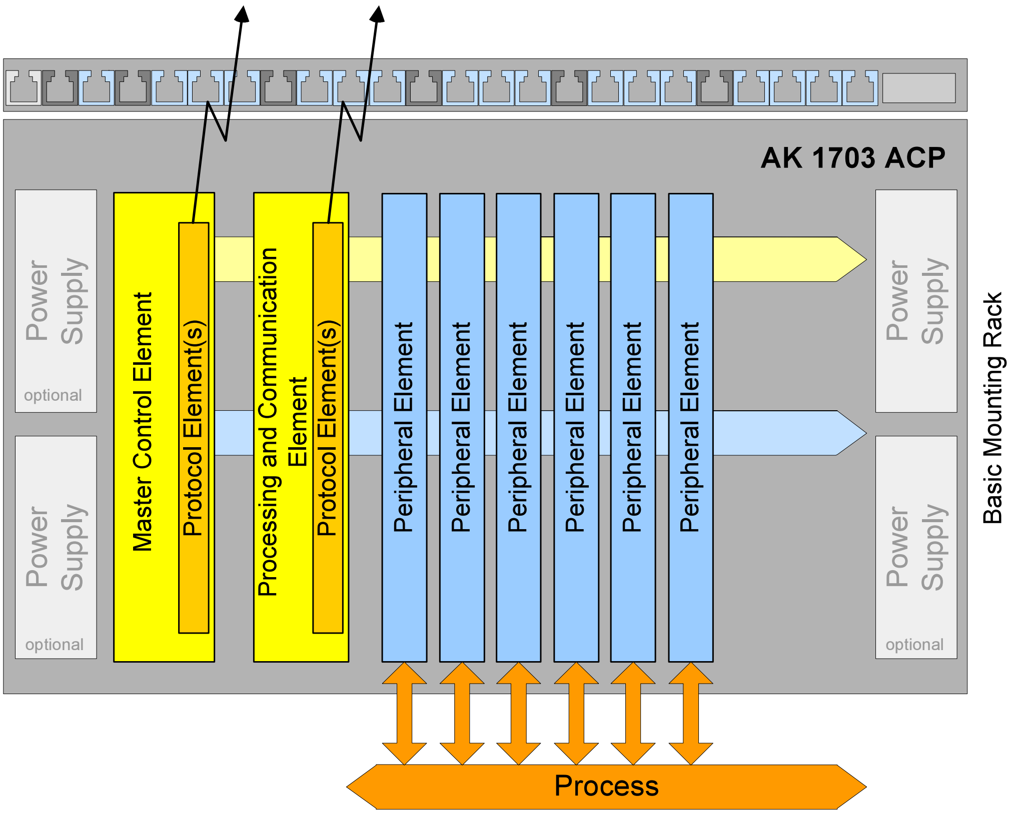
RJ45 Connection Technique for external AX 1703 Peripheral Bus



3.2. System Architecture

3.2.1. Configuration

3.2.1.1. Basic Mounting Rack AK 1703 ACP



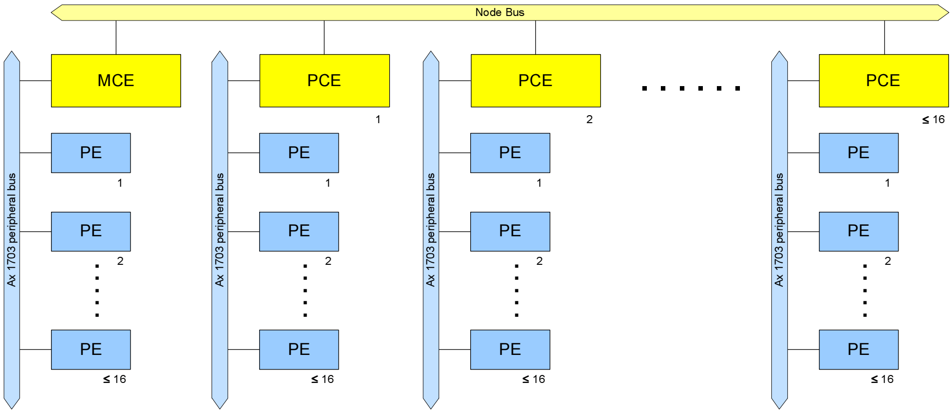
An AK 1703 ACP forms an automation unit of the system family SICAM 1703 and is structured from the following elements:

- Master control element (*)
- Processing and communication element(s) (*)
- Protocol element(s) (*)
- Peripheral element(s) (*)
- Mounting rack with one to four power supplies

Parts that are marked with (*) are System Elements.

Peripheral elements can also be installed outside of the basic mounting rack.

3.2.1.2. Configuration of an Automation Unit AK 1703 ACP



Legend:

MCE Master control element
PCE Processing and communication element
PE Peripheral element

Configuration

- 1 master control element
  - forms the independent Ax 1703 peripheral bus *AXPE-C0*, on which up to 16 peripheral elements can be installed
- up to 16 processing and communication elements
  - each processing and communication element forms an independent Ax 1703 peripheral bus *AXPE-Cn* (n = 1..16), on which up to 16 peripheral elements can be installed
- up to 272 peripheral elements
- up to 66 protocol elements (interfaces with individual communication protocol)
  - up to 2 protocol elements on the master control unit
  - up to 4 protocol elements for each processing and communication element

Ax 1703 peripheral bus

The *AXPE-Cn* buses (n = 1..16) are available on connectors, in order to be able to connect peripheral elements outside the basic mounting rack. One of the buses can be selected, in order to also supply those peripheral elements in the basic mounting rack.

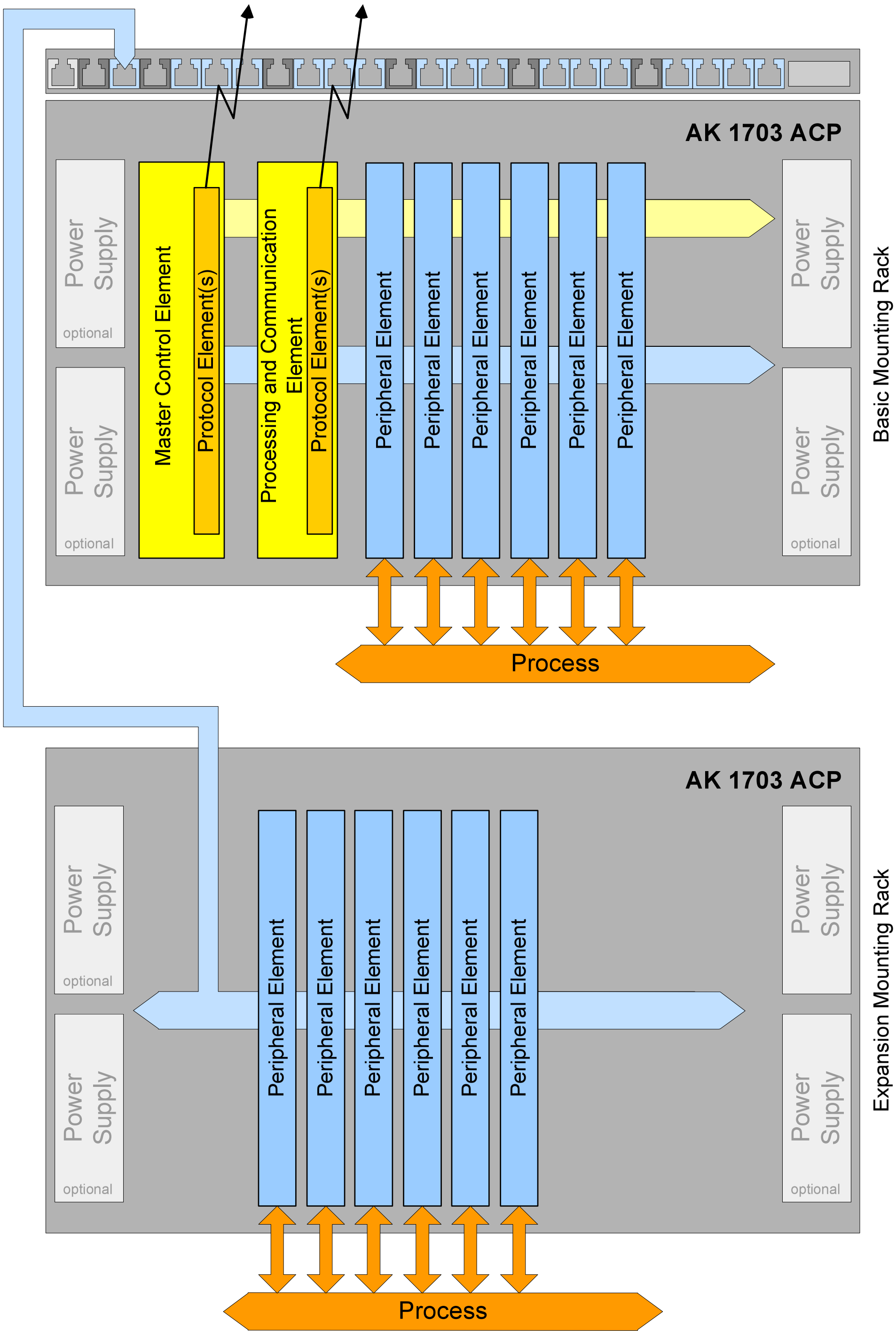
3.2.1.3. Peripheral Elements outside the Basic Mounting Rack

Peripheral elements may be

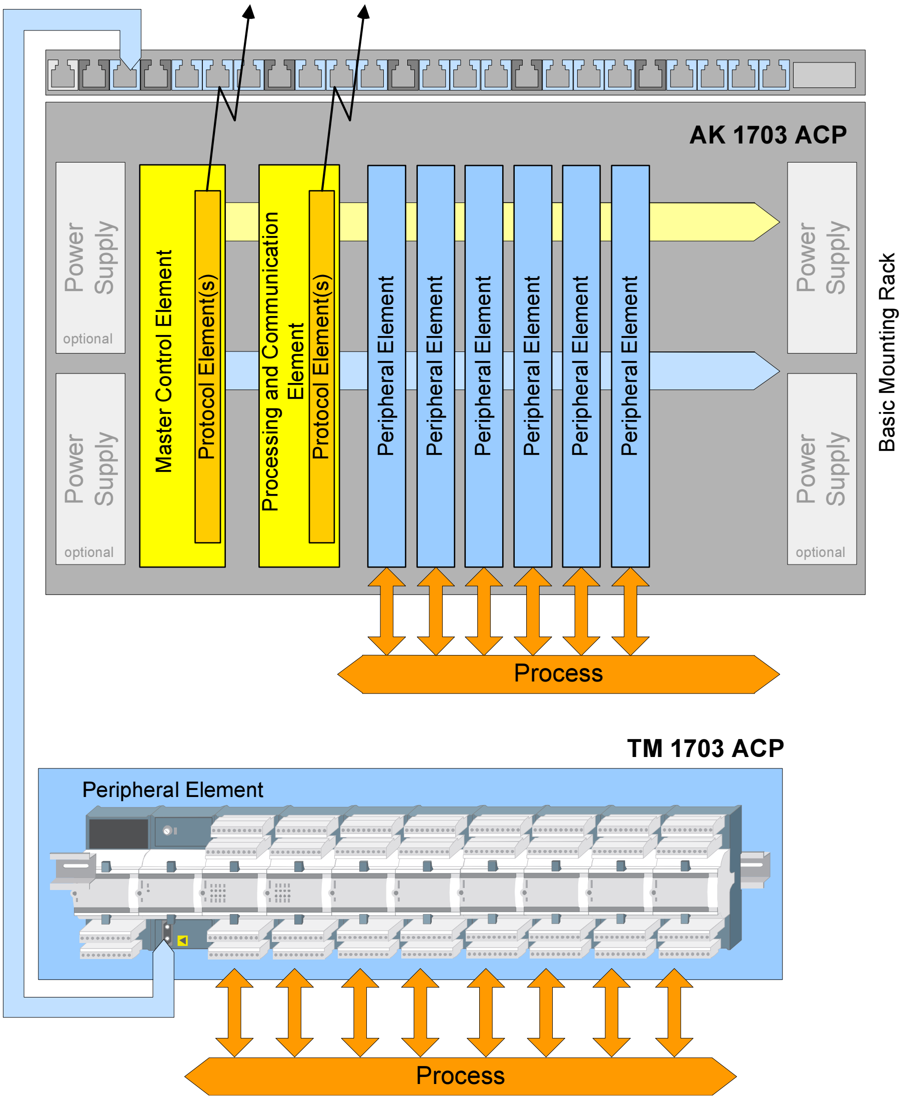
- installed in the basic mounting rack (as shown above), and/or
- installed in an expansion mounting rack and connected electrically, and/or
- installed at remote locations and connected electrically or optically

Basic examples with peripheral elements from the system families AK 1703 ACP and TM 1703 ACP can be found below.

3.2.1.3.1. Basic Mounting Rack + Expansion Mounting Rack



3.2.1.3.2. Basic Mounting Rack + TM 1703 ACP Peripheral Element

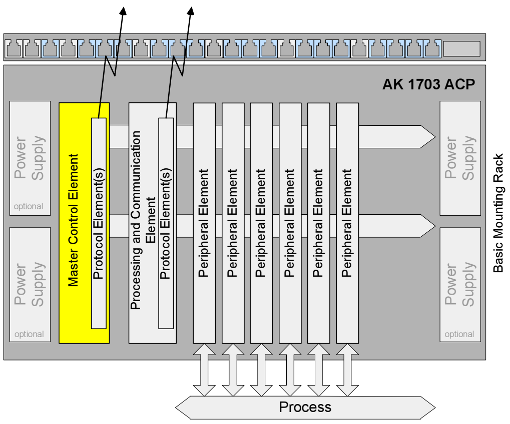


Comprehensive information on configuration and connection of peripheral elements can be found in the document ACP 1703 Platforms - Configuration Automation Units and Automation Networks.

3.2.2. System Elements

A system element is a functional unit and consists of hardware and firmware. The firmware gives the hardware the necessary functionality.

3.2.2.1. Master Control Element

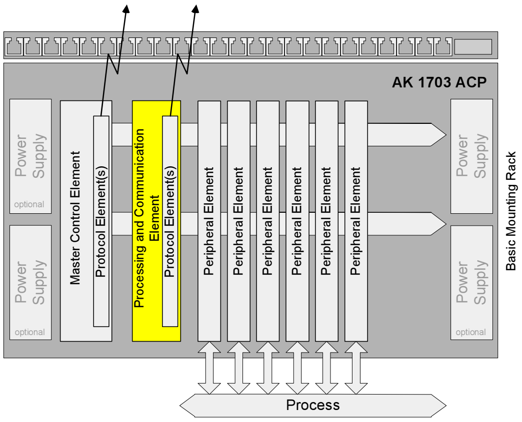


The master control element forms the heart of the automation unit.

Functions of the master control element:

- Communication with installed peripheral elements via the serial Ax 1703 peripheral bus
- Open-/closed-loop control function with a freely programmable user program according to IEC 61131-3, e.g. in function diagram technology
- Communication with other automation units via protocol elements installable on the master control element (up to 2 interfaces)
- Central coordinating element for all system services and all internal and overlapping concepts, such as e.g.
  - Data Flow Control
  - Monitoring functions
  - Diagnostic
  - Time management and time synchronization via minute pulse, serial time signal (DCF77/GPS-receiver), serial communication link, NTP - Server over LAN/WAN
  - Local TOOLBOX II connection
  - Storage of parameters and firmware on a Flash Card

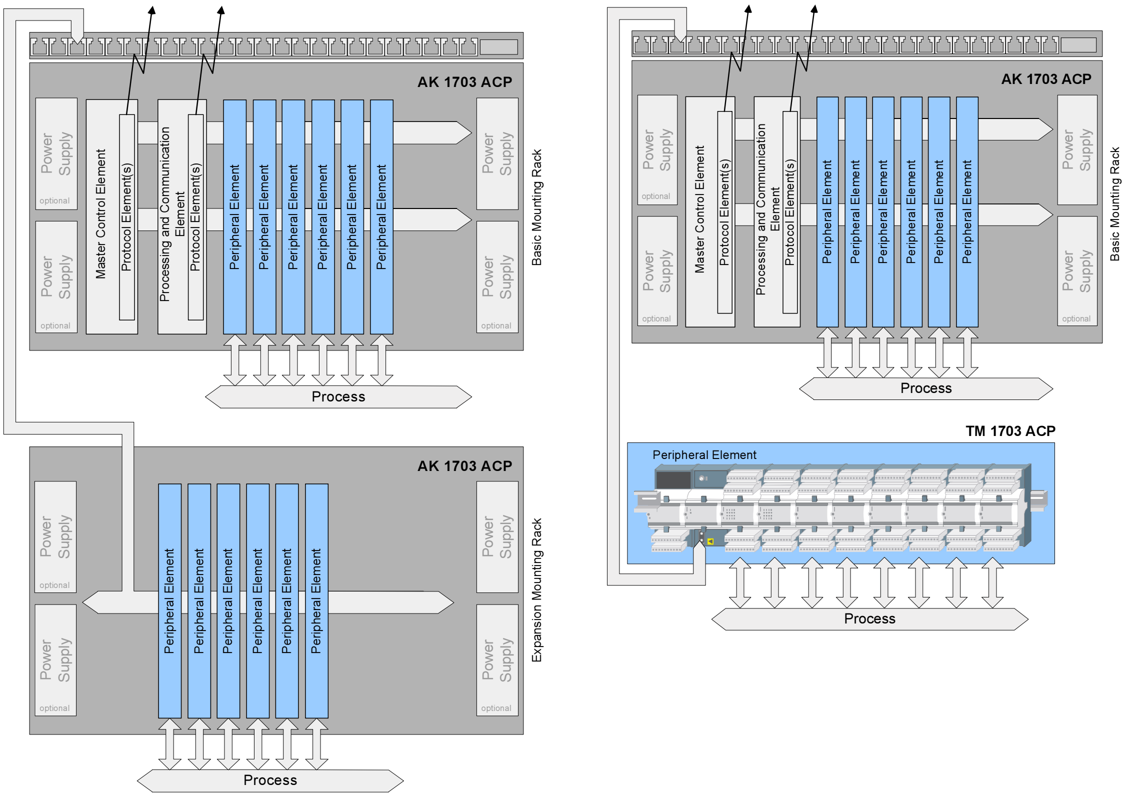
3.2.2.2. Processing and Communication Element



Functions of the processing- and communication elements:

- Communication with installed peripheral elements via the serial Ax 1703 peripheral bus
- Open-/closed-loop control function with a freely programmable user program according to IEC 61131-3, e.g. in function diagram technology
- Communication with other automation units via protocol elements installable on the processing and communication element (up to 4 interfaces)

3.2.2.3. Peripheral Element



Peripheral elements are used for acquisition or output of process information and perform process-oriented adaption, monitoring and processing of the process signals at each point where the signals enter or leave an automation unit. Processing is performed to some degree by

- hardware (e.g. filter, ADC, DAC) and by
- firmware (e.g. smoothing of measured values, time tagging)

The peripheral elements deliver over the Ax 1703 peripheral bus

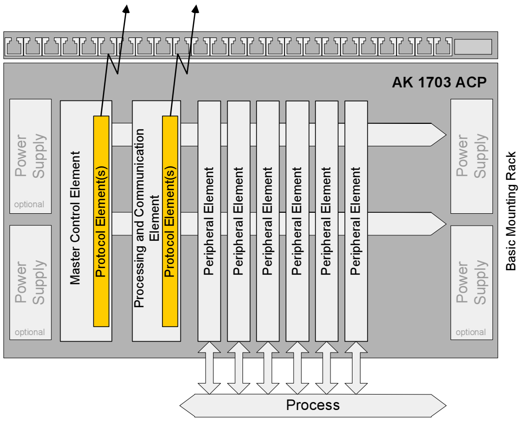
- periodical information,
- messages with process information, and
- messages with system information (e.g. diagnostic information)

and receive

- messages with process information,
- periodical information, and
- messages with system information (e.g. parameters)

Comprehensive information on configuration and connection of peripheral elements can be found in the document ACP 1703 Platforms - Configuration Automation Units and Automation Networks.

3.2.2.4. Protocol Element



A protocol element is used for the exchange of data - and thereby for the transmission of messages - over a communication interface to other automation units or devices of third-party manufacturers, e.g. control systems.

The hardware of a protocol element is a communication interface which - dependent on system and interface - can be available in different ways:

- integrated on a basic system element
- on a serial interface module (SIM), which is installed - directly or cascaded (SIM on SIM) - on the basic system element

In order to be able to communicate with as many systems as possible, SIEMENS has decided not only to employ standard protocols, which are normalized by IEC, but to considerably take part in the creation of these standards. Therefore, the automation unit communicates externally as well as internally according to these standards:

- IEC 60870-5-101 in case of serial communication
- IEC 60870-5-103 in case of serial communication with protective devices
- IEC 60870-5-104 in case of LAN/WAN networks

On every interface provided by the SIM, a communication protocol available for the interface can be loaded with the TOOLBOX II.

3.2.2.4.1. Serial Communication

For serial communication are available as standard protocols:

- point-to-point traffic
- multi-point traffic, optionally with relay operation
- dial-up traffic

Naturally, all standard protocols are fully based on the interoperable standard to IEC 60870-5-101/103, including

- absolutely free addressing
- single object orientation
- time Synchronization
- integrated remote maintenance functions such as
  - remote diagnostics
  - remote parameter setting
  - online test functions

Yet, there is still a whole series of other available protocols such as:

- counter interfacing according to IEC 61107
- interfacing of protective devices according to IEC 60870-5-103
- Modbus
- DNP 3.0

Additional information on interfacing to non-SIEMENS systems and third-party protocols (protocols subject to license) is available on request.

3.2.2.4.2. LAN / WAN - Communication

Today, modern automation systems are generally distributed and thus require networks to connect the various components with one another. In its systems, SIEMENS has for many years provided networks under the SATNET name, which link the various components with one another.

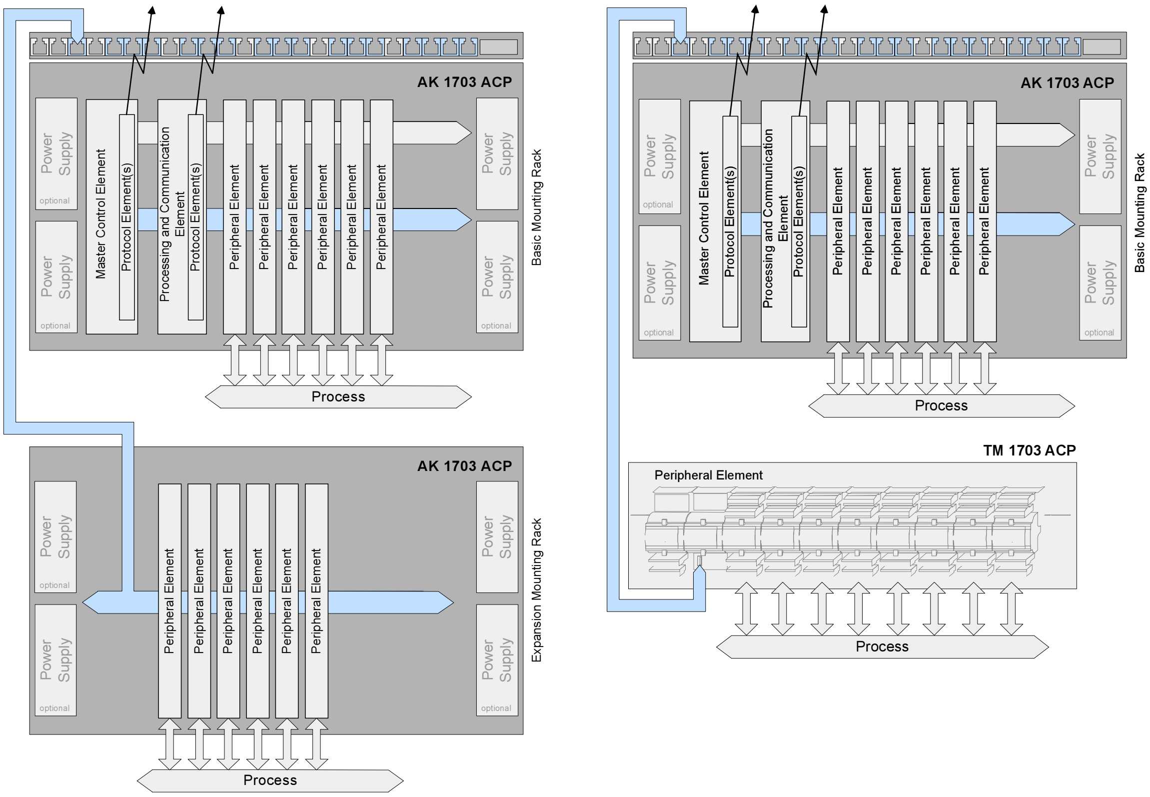
From the very beginning, great attention was paid to ensuring full integration as well as optimum availability and operational reliability. As network technology continued to become ever more refined, SATNET, as well, has continuously been updated and upgraded to reflect the latest state of the art without neglecting the criteria of ensuring a long system lifecycle and highest availability.

LAN/WAN communication relies on Ethernet TCP/IP to IEC 60870-5-104, which again guarantees maximum interoperability.

If AK 1703 ACP is used as a station control device, then the communication with the devices of the bay level (bay control units, protective devices) can also be established according to IEC 61850.

IEC 61850 is the standardized communication standard for substation automation which interconnects devices of the bay level and the station control level, based on Ethernet TCP/IP.IEC 61850 is the standardized communication standard for substation automation which interconnects devices of the bay level and the station control level, based on Ethernet TCP/IP.

3.2.3. Ax 1703 Peripheral Bus



The Ax 1703 peripheral bus permits the secured (hamming distance 4), serial system-internal communication between the basic system element and the peripheral elements.

Depending on the system, the Ax 1703 peripheral bus may be accessible:

- at backplane slots, and/or
- via external connectors (optical or electrical).)

Serial communication also makes it possible to locally detach individual or all peripheral elements without sacrificing any of the full system functionality - with an optical connection up to 200m, with an electrical connection up to 3m.

The communication at the Ax 1703 peripheral bus takes place according to the master-slave method, the peripheral elements being the slave and the basic system element the master.

Each peripheral element constitutes, regardless of its

- function,
- data volume,
- processing,

one participants at the Ax 1703 peripheral bus.

Addressing of the bus participants is handled for all peripheral elements, whose hardware is not integrated into the hardware of a basic system element, via a logical peripheral board address (PBA) that can be set on the peripheral element. Integrated peripheral elements have a pre-defined PBA.

Data of different classes are transmitted over the Ax 1703 peripheral bus:

- periodical data for the function package Automation, and
- spontaneous data for the function package Telecontrol.

Periodical information is exchanged between the basic system element and the peripheral element based on the cycle of the open-/closed-loop control function on the basic system element.

Spontaneous data are transmitted as messages with process- and system information between the basic system element and the peripheral element, with acknowledgement.

Hint

In order to connect peripheral elements electrically and/or optically, resources such as bus interface modules may be required (see ACP 1703 Platforms - Configuration Automation Units and Automation Networks).

3.3. System Elements in AK 1703 ACP

Master Control Element

	Designation	
CP-2010/CPC25	System Functions, Processing and Communication	required

Processing and Communication Element

	Designation	
CP-2012/PCCE25	Processing & Communication	optional
CP-2017/PCCX25	Processing & Communication	optional

Peripheral elements AK 1703 ACP

	Designation	
DI-2100/BISI25	Binary signal input (8x8, 24-60VDC)	optional
DI-2110/BISI26	Binary signal input (8x8, 24-60VDC)	optional
DI-2111/BISI26	Binary signal input (8x8, 110/220VDC)	optional
DO-2201/BISO25	Binary Output (Transistor, 40x1, 24-60VDC)	optional
AI-2300/PASI25	Analog in-/output (16x ±20mA + 4x2 opt. expans.)	optional
DO-2210/PCCO2X	Checked command output 24-60 VDC	optional
MX-2400/USIO2X	Signal Input/Output (24-60VDC, +/-20mA, opt. exp.)	optional

Supported peripheral elements TM 1703 ACP

	Bezeichnung	
PE-6400/TCIO65	Peripheral Controller for TC 1703 (Ax-PE bus el)	optional
PE-6401/TCIO65	Peripheral Controller for TC 1703 (1x Ax-PE bus opt)	optional
PE-6402/TCIO65	Peripheral Controller for TC 1703 (2x Ax-PE bus opt)	optional
PE-6400/USIO65	Peripheral Controller (Ax-PE bus el)	optional
PE-6401/USIO65	Peripheral Controller (1x Ax-PE bus opt)	optional
PE-6402/USIO65	Peripheral Controller (2x Ax-PE bus opt)	optional
PE-6410/TCIO66	Peripheral Controller for TC 1703 (Ax-PE bus el)	optional
PE-6411/TCIO66	Peripheral Controller for TC 1703 (1x Ax-PE bus opt)	optional
PE-6412/TCIO66	Peripheral Controller for TC 1703 (2x Ax-PE bus opt)	optional
PE-6410/USIO66	Peripheral Controller (Ax-PE bus el)	optional
PE-6411/USIO66	Peripheral Controller (1x Ax-PE bus opt)	optional
PE-6412/USIO66	Peripheral Controller (2x Ax-PE bus opt)	optional

Supported peripheral elements AM 1703

	Bezeichnung	
DI-1112/BISI15	Binary Signal Input (3x8, 24-60VDC)	optional
DI-1113/BISI15	Binary Signal Input (3x8, 110/220VDC)	optional
AI-1304/TIPP16	Direct Transformer Input (4x220V,3x6A)	optional
MX-1416/USIO15	Signal Input/Output (DI:24-60VDC, I+U, DO:30A)	optional
MX-1417/USIO15	Signal Input/Output (DI:110/220VDC, I+U, DO:30A)	optional

Information about a system element can be found in the relevant data sheet (see "Literature").

Protocol Elements

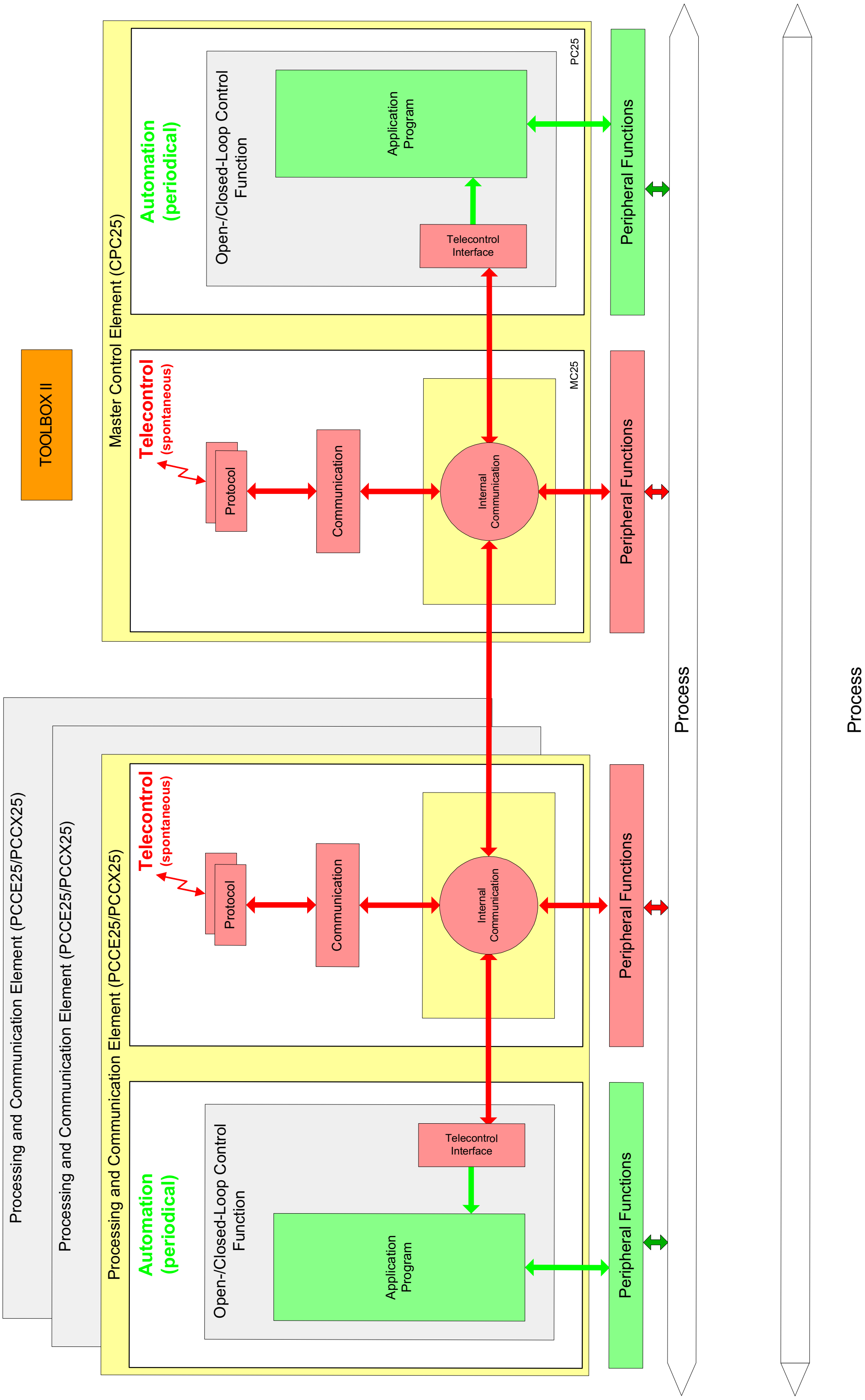
	Bezeichnung	
SM-X551/BPPA0	Standard protocol for point-to-point traffic	optional
SM-X551/UMPMA0	Standard protocol for multi-point traffic (M)	optional
SM-X551/UMPSA0	Standard protocol for multi-point traffic (S)	optional
SM-X551/SFBMA1	Standard protocol for field bus (M)	optional
SM-X551/SFBSA1	Standard protocol for field bus (S)	optional
SM-X551/DIAMA0	Standard protocol for dial-up traffic (M)	optional
SM-X551/DIASA0	Standard protocol for dial-up traffic (S)	optional
SM-X551/103MA0	Standard protocol for interfacing of protective devices (M)	optional
SM-2541/BPP00	Standard protocol for point-to-point traffic	optional
SM-2541/UMPM02	Standard protocol for multi-point traffic (M)	optional
SM-2541/UMPS00	Standard protocol for multi-point traffic (S)	optional
SM-2541/UMPM01	Standard protocol for field bus master (M)	optional
SM-2541/UMPS01	Standard protocol for field bus master (S)	optional
SM-2541/DIAM00	Standard protocol for dial-up traffic (M)	optional
SM-2541/DIAS00	Standard protocol for dial-up traffic (S)	optional
SM-2541/103M00	Standard protocol for interfacing of protective devices (M)	optional
SM-2545/DPM00	Standard protocol for Profibus DP	optional
SM-2554/ET02	Standard protocol for Ethernet TCP/IP IEC104	optional
SM-2554/ET03	Standard protocol for Ethernet TCP/IP IEC61850	optional
SM-2556/ET02	Standard protocol for Ethernet TCP/IP IEC104	optional
SM-2556/ETA2	Standard protocol for Ethernet TCP/IP IEC104	optional
SM-2557/ETA2	Standard protocol for Ethernet TCP/IP IEC104	optional
SM-2556/ET03	Standard protocol for Ethernet TCP/IP IEC61850	optional

(M) ... Master (S) ... Slave

Information on a protocol element can be found in the relevant data sheet (see "Literature").

Subject-to-license protocols on request.

3.4. Firmware Architecture



3.5. Engineering

The costs for the creation and maintenance of automation technology plants are determined to an increasing degree by the costs for the creation and updating of the engineering data. The engineering data therefore represent major capital goods of the company, the creation and updating of which by means of a high-quality engineering system results in a considerable reduction of the indirect costs.

For this reason, SIEMENS places great importance on the engineering systems in its product range, and with the TOOLBOX II, thus consequently continues its policy of always providing high-quality, ergonomic products based on innovative system technology, also in the field of engineering systems.

The high demands on the easy and intuitive operability as well as on the overall ergonomics of an engineering system are satisfied by the TOOLBOX II through an operating and display technology based on the latest state of technology.

- Fully graphical user interface with easy operation and uniform "Look and Feel" (Window technology, Menus, Icons, Help System)

as well as through forward-looking conception

- Industrial standard database system ORACLE ™,
- Network support (TOOLBOX II Peer Server),
- Client/Server architecture
- Standard operating systems (Windows NT, Windows 2000, Windows XP)
- Standard hardware (Personal Computer)

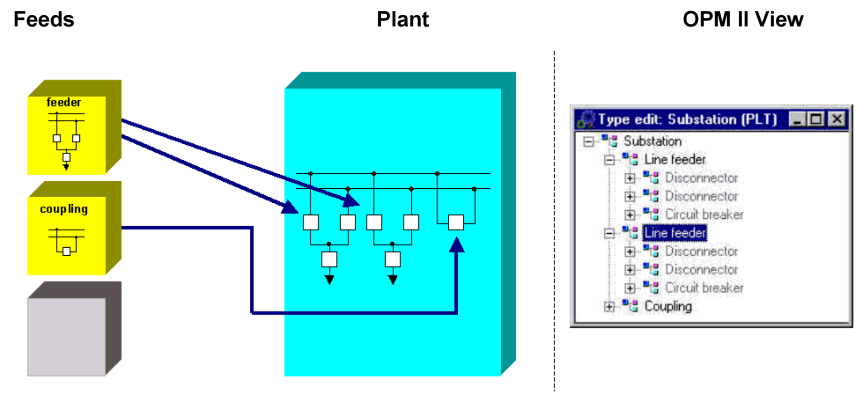
The TOOLBOX II constitutes an integrated overall tool

- all engineering data are stored, managed and processed – on the same hardware platform in a central database – by the same tool; a transfer of data between devices and tools is not necessary
- besides the entire SICAM 1703 product line, SAT 200 is also configured by means of the TOOLBOX II
- if an exchange of data with devices or systems of other manufacturers should be necessary, then there is a specific data interface for this available that is simple to operate, the source data management

During the entire lifecycle of the plant, the TOOLBOX II comprehensively supports all phases of the plant configuration and maintenance for the entire SICAM 1703 system family. The engineering with the TOOLBOX II therefore goes far beyond conventional device parameterization and comprises the following areas:

- Data Acquisition, Data Modeling
- Parameter setting, test and diagnostic
- Documentation
- Backup and archiving
- Maintenance

3.5.1. Data Acquisition, Data Modeling



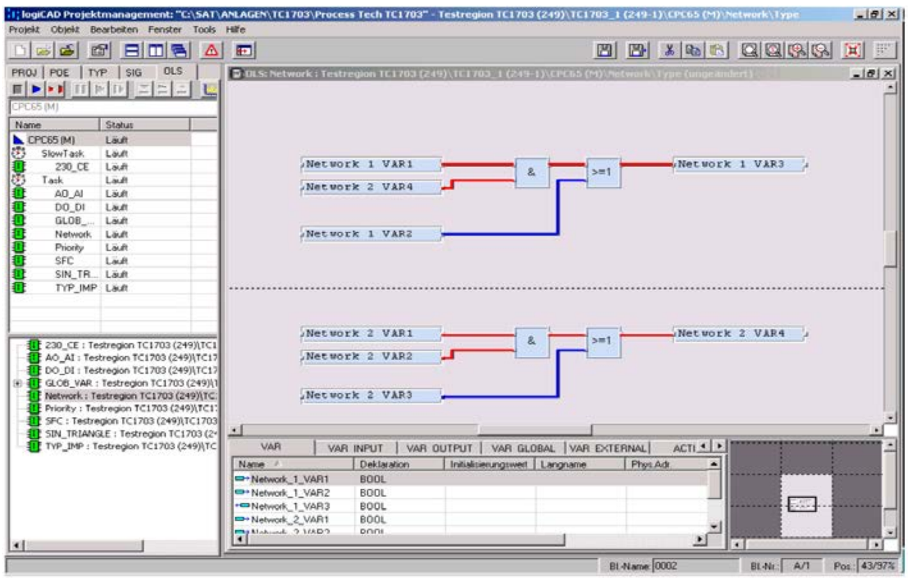
The plant is at the center of the configuration procedure

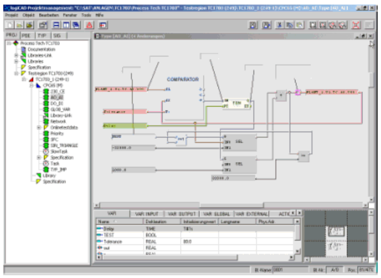
For <u>same type</u> primary technology units and resources (objects), types can be modeled, which contain the characteristics of the objects. During engineering, such objects can be created identically many times very easily (one speaks of instantiation). As a result considerable savings can be achieved.

The advantages outweigh not only with the creation but also with the updating of the data for expansions and with consistent change of all same-type objects as well as with the achievable quality with regard to consistency of the engineering data.

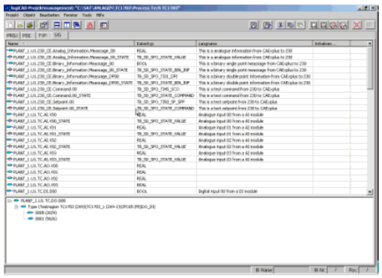
3.5.2. User Programs for the Open-/Closed-Loop Control Function

User programs for the open-/closed-loop control function are created according to IEC 61131-3 using **CAEx *plus***, a tool of the TOOLBOX II. This standard is generally accepted in the market and is recognized. Engineering according to this standard generally only requires short training periods.

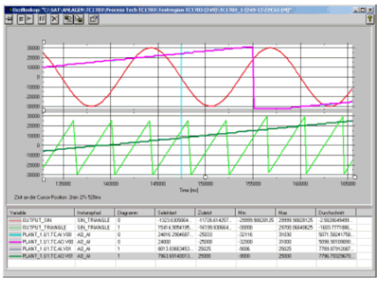




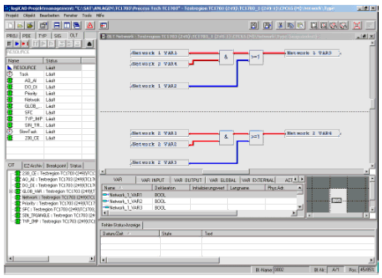
Function block diagramm-editor



Signal list



Oszilloscope function



Online test

Thereby versatile applications can be easily realized:

- Logical links
- Sum commands, sum alarms
- Limit monitoring
- Bay- and station-related interlocks
- Synchronous comparison with analog of busbar
- Switchover automation, switching sequences (busbars, transformers etc.)
- Step-by-step controls
- Closed loop control (tap changer controller, turbine governor, ...)

There are many test function available, both offline as well as online. Consequently, test and commissioning periods can be kept very short.

4. Function Packages

Contents

4.1. Telecontrol

The function package Telecontrol includes the following functions:

- Process input and output on peripheral elements
- Communication with other automation units
- Protocol elements
- Automatic data flow routing
- Data storage
- Priority control
- Redundant communication routes
- Communication within the automation unit
- Protocol element control and return information

4.1.1. Process Input and Output on Peripheral Elements

- Acquisition of the process data on the peripheral elements, processing, generation and transfer of *messages with process information* (time tag with 1ms or 10ms resolution, dependent on the respective peripheral element) over the Ax 1703 peripheral bus
- Reception of *messages with process information* over the Ax 1703 peripheral bus, processing and output of the process data

Hint

These functions are described in detail in the document *SICAM 1703 Common Functions Peripheral Elements according to IEC 60870-5-101/104* (refer to Further Documents).

4.1.2. Communication with other Automation Units

The communication function controls the transmission of messages via protocol elements to other automation units or control systems.

A protocol element is based on hardware integrated in a basic system element or on a serial interface module (SIM) that can be installed on a basic system element for serial, LAN/WAN and field bus communication and supports standard protocols according to IEC 60870-5-101/103/104 and a large number of protocols for the communication with third-party systems.

The communication function differentiates between transmission and receive direction.

Communication function in transmit direction

The messages to be transmitted are learned through the automatic data flow routing and stored in the data storage. The transfer of the messages from the data storage to the protocol elements takes place via a priority controller in order to optimally utilize the transmission route.

- Automatic data flow routing
- Data storage
- Priority control

Communication function in receive direction

- *Messages with process information* are distributed to all functions within the automation unit.
- *Messages with system information* are either processed directly (e.g. station interrogation) or distributed further based on their destination address (CASDU) (e.g. messages for remote maintenance).

4.1.2.1. Automatic Data Flow Routing

For the data flow routing, a routing of individual process information items is not necessary. Simply only the direction (monitor direction, control direction, both directions), in which the messages are to be transmitted, is to be parameterized.

The type identification of each message provides information about the class (refer to Messages with Process Information) to which a message belongs and with which methods it is to be distributed:

- *Messages with process information in monitor direction*
  - In simple applications, the messages are distributed via an entry in the topology.
  - For more complex applications, the messages can be distributed selectively with the help of the *data flow filter*:

    For each communication interface, pass-through filters or blocking filters can be set. Since wildcards can also be used for all address attributes of the message, it is possible to control the data flow very specifically with simple means.

- *Messages with process information in control direction*
  - The messages are distributed to the destinations determined by their CASDU over interfaces that are defined in the topology. The CASDU is interpreted as destination address.

4.1.2.2. Data Storage

The messages that are intended for transmission over communication interfaces, are in principle stored chronologically in rings. There is a process image both before and after a ring. The arrangement, consisting of one ring and two process images, is called a priority channel (priority channels for transparent data do not have any process images).

Depending on the data communication mode of the protocol element over which the communication is processed, priority channels are provided for every priority of the messages to be transmitted and for every station that can be reached via the protocol element:

- Data communication mode "Multi Point" (e.g. multi-point traffic, LAN)

  One priority channel for every transmission priority, for every station and for every protocol element

- Data communication mode "Single Point"

  One priority channel for every transmission priority and for every protocol element

With regard to the data that they transport, priority channels are distinguished as follows:

- Time synchronization
- System information
- Process information in control direction
- Process information in monitor direction Priority HIGH with class 1 data
- Process information in monitor direction Priority MEDIUM with class 2 data
- Process information in monitor direction Priority LOW with class 2 data
- Transparent information

Functions for priority channels:

- State compression for measured values (can be set using parameters)

  Specifically reduces the flood of messages, that can continuously generate fluctuating ("floating") measured values

- Behavior with a priority channel overload
- Behavior during a communication failure (transmit direction)
- Monitoring of the dwell time (parameter-settable) of *messages with process information in control direction*

  Messages that are stored too long in the priority channel are discarded

- Answering of station interrogations
- Behavior during failure of peripheral elements, communication interfaces etc.
- Blocking (series of information elements)

4.1.2.3. Priority Control

The priority controller has the task of selecting messages recorded in the data memories independently and individually for each interface and station and to direct the transmission of the messages via the protocol elements in accordance with their priority. This ensures, that with several informations queued at the same time, the higher-priority, highly important information is transmitted first.

The prioritization does not however represent an absolute priority status, but rather a measure for dividing up the channel capacity. This ensures, that even with continuously available higher-priority data, those of lower priority can also be transmitted.

4.1.2.4. Redundant Communication Routes

In accordance with the requirements with regard to reaction time, availability, data throughput and transmission media, the following redundancy operating modes are possible:

- communication with redundant remote stations
- redundant communication with a remote station (load share operation)

Communication with redundant remote stations

The communication with redundant remote stations is possible with every data communication mode (single point, multi point). Thereby both transmission paths are operated independent of each other.

Redundant communication with a remote station

The redundant communication to a remote station is enabled through two operating modes:

- **Data Split Mode**
  - improvement in the availability
  - different baud rates possible on both transmission paths
  - with a failure of one interface, a switchover to the other interface is performed, the data throughput remains unchanged at the same baud rates

- **Load Share Mode**
  - improvement of the data throughput <u>and</u> the availability through optimum utilization of the interfaces with every load and in every operating state
  - same baud rates recommended on both transmission paths
  - with a failure of one interface, from that point on the entire traffic is processed over the available interface; the data throughput sinks to that value, that would otherwise be achieved with the same conditions in data split mode

The redundant communication to a remote station

- is supported
  - exclusively in the data communication mode "single point"
  - only on the two interfaces of a serial interface module.

4.1.3. Communication within the Automation Unit

Within an automation unit, the function package Telecontrol communicates with

- the function package Automation via its Telecontrol Interface

The functions are described there.

4.1.4. Protocol Element Control and Return Information

This function is used for the user-specific influencing of the functions of the protocol elements. The main application lies with protocol elements with multi point data communication mode and especially for dial-up traffic configurations.

This function contains two separate independent parts:

- Protocol element control
  - test if stations are reachable
  - suppression of errors with intentionally switched-off stations

- Protocol element return information
  - cost control of telephone charges
  - cost-efficient utilization of the telephone line
    (e.g. command initiation only then, when a connection has already been established).

4.2. Automation

The function package Automation includes the following functions:

- Process input and –output on peripheral elements
- Telecontrol functions
  - Treatment for commands according to IEC 60870-5-101/104
  - Change monitoring and generation of messages with time tag
- Open-/Closed-loop control function

4.2.1. Process Input and Output on Peripheral Elements

- Acquisition of the process data on the peripheral elements, processing, periodical transfer of the process information to the open-/closed-loop control function
- Periodical acceptance of process information from the open-/closed-loop control function, processing and output of the process data

Hint

These functions are described in detail in the document *SICAM 1703 Common Functions Peripheral Elements according to IEC 60870-5-101/104* (refer to Further Documents).

4.2.2. Open-/Closed-Loop Control Function

The open-/closed-loop control function is used for the management of automation tasks with the help of an application program.

The creation of the application program is carried out by the TOOLBOX II with the tool **CAEx *plus*** predominantly in function diagram technology according to **IEC 61131-3.**

The application program processes process information (so-called signals) from the peripheral elements connected to the basic system element and / or from other system elements in the automation network of the specific process-technical plant.

Process images form the interface of the application program to the outside world. We distinguish between input process images and output process images.

The exchange of the process information can take place in two ways:

- Transmission of periodical information from and to the peripheral elements connected to the basic system element via the Ax 1703 peripheral bus (Process Input and Output on Peripheral Elements)
- Transmission of spontaneous information objects from and to functions or peripheral elements within the automation unit, other open-/closed-loop control functions and other automation units or control systems with the help of the telecontrol function (Transfer of Messages with Process Information and Change Monitoring and Generation of Messages with Time Tag)

The open-/closed-loop control function supports 32 programs (type instances), whereby each program is assigned one of three periodical tasks. As a result, fast controls can be optimally combined with slower background processings.

The management of these three periodical tasks (Task Management) corresponds with the standard IEC 61131-3. Spontaneous tasks are not supported.

Variables, signals (input process images for spontaneous information objects) and function blocks can be saved non-volatile. That means, that after a power failure these variables and signals are immediately available again with their values before the power failure.

4.2.2.1. Telecontrol Interface

4.2.2.1.1. Transfer of Messages with Process Information

Reception of *messages with process information* and transfer to the open-/closed-loop control function for the purpose of further processing:

Messages with process information in monitor direction:

- Single-point information, double-point information, step position information
- Measured values
- Integrated totals
- Bitstring of 32 bit

Messages with process information in control direction:

- Single commands, double commands, regulating step commands
- Setpoint commands
- Bitstring of 32 bit

Treatment for commands according to IEC 60870-5-101/104

The treatment for commands serves for the check of the spontaneous information objects to be processed with the help of the open-/closed loop control function and transmission of the confirmation for:

- Pulse commands (single commands, double commands, regulating step commands)
- Setpoint values (setpoint command)
- Bitstring of 32 bit

The data transfer of the spontaneous information objects to the application program of the open-/closed-loop control function for further processing is dependent on the result of the checks.

The activation of the element or function to be controlled is the task of the application program of the open-/closed-loop control function.

For the proper operation of this function, information is required by the application program of the open-/closed-loop control function (e.g. from an interlocking logic) for the choice of a positive or negative confirmation.

The treatment for commands can be activated individually for each command via a parameter.

The treatment for pulse command comprises the following processing functions:

- prepare command output procedure
  - formal check
  - retry suppression
  - 1-out-of-n check
  - direct command or
    select and execute command
  - control location check
  - command locking
  - system-element overlapping 1-out-of-n check
- initiate command output procedure
  - command to application program
- monitor pulse duration (only pulse commands)
  - command output time
  - return information monitoring
- terminate command output procedure

4.2.2.1.2. Change Monitoring and Generation of Messages with Time Tag

For the generation of messages with process information, the signals in the output process images that are assigned to an element of a spontaneous information object, are monitored for change.

The change monitoring takes place in a grid of the cycle time of each task, in which the signal is assigned to a spontaneous information object.

On a change of the state in a corresponding element of the spontaneous information object, the generation of the message is initiated.

Depending on the type of signal to be monitored, different methods are applied:

- Change of the state (positive edge, positive and negative edge)
- Change of the value (according to the rules of the additive threshold value procedure)

If a spontaneous information object has been activated for transmission due to a change, a *message with process information* is generated. The time tag represents either the current time synchronous with the cycle (resolution 10ms or multiples of 10ms) or the time information from an assigned spontaneous information object.

Additive threshold value procedure

The additive threshold value procedure prevents an unnecessary loading of the transmission links with insignificant changes of the corresponding signal and acts only on the basic data of the spontaneous information objects with measured values.

4.2.2.2. Task Management

The open-/closed-loop control function manages the application programs in 3 tasks running periodically:

- "Fast Task"
- "Task"
- "Slow Task"

Coordination of the sequences of a task

- Periodical start in the selected cycle
- Input handling
- Program processing
- Output handling
- Online Test
- Real time archive

Coordination of the three tasks with each other

- "Fast Task" runs without interruption and with constant running time
- "Task" and "Slow Task" can be interrupted by higher-priority functions

Cycle Time

- Within the cycle time, all programs assigned to a task (type instances) must process the input handling and the output handling for this task
- The cycle time can be set in the tool **CAEx *plus*** for each task.
- The cycle times of the three tasks must be different and ascending from the "Fast Task" to the "Slow Task".

Watchdog Timer

This function monitors the proper sequence of each task within its set cycle time. If a task is not finished with its input handling, program processing and its output handling within this time, the next cycle for this task is omitted and a time-out is signaled.

With serious time-outs, for example due to a malfunction, the reliability of the application program becomes questionable. A time scale can be defined for such cases, the exceeding of which leads to an error message and a controlled shutdown of parts or the entire application program as well as all peripheral elements connected.

4.2.2.3. Loading the Application Program

Initial Loading

The initial loading of an application program is always associated with a startup.

Loading of changes (Reload)

Frequently, in the test and commissioning phase but also with the remedy of faults, changes must be carried out. Most such changes (error rectifications, expansions) to the application program can be loaded without interruption to operation. Far-reaching changes can necessitate a startup of the basic system element and consequently an interruption to operation.

In the case of a loading operation that does not necessitate any interruption to operation, all tasks of the open-/closed-loop control function continue to run unaffected. After successful loading, a switchover to the newly loaded application program takes place synchronous with the cycle.

Examples of changes that do not necessitate any interruption to operation:

- if after change, the function corresponds completely with that before change, in other words a change has been performed that is not noticeable from "outside"
- if only new functional parts were added, that do not affect those that already existed
- if parameters of a controller are adapted

Fundamentally however, the fault-free operation and consequently the availability of every control or controller depends on the quality of the program – in other words the measure of how free they are of formal and logical errors. The loading of error-burdened changes can always lead to interruptions to operation.

4.2.2.4. Online Test

The entire functionality of the Online Test applies to

- the TOOLBOX II tool "CAEx *plus* Online Test" and
- the Online Test function of the open-/closed-loop control function of the automation unit

While in the tool "CAEx *plus* Online Test", all functions of the man-machine-interface can be found, the open-/closed-loop control function provides functions for the execution of the operator inputs.

If for example a value is to be displayed, then the selection of the value and its display takes place in the tool "CAEx *plus* Online Test". For this purpose, the open-/closed-loop control function is given the task of reading out the selected value and transmit it to the TOOLBOX II.

In the following, those functions are listed that the Online Test function of the open-/closed-loop control function provides.

Display and setting of variables and signals

- Display of variables and signals
- Single setting of variables (Single Forcing)

  The value of a variable can change again at any time after setting, due to the function of the program
- Permanent setting of variables and signals (Permanent Forcing)

  In order to be able to set variables and signals permanently, a special element ("Force Marker") is set in the function diagram. This element contains the set value and a switch. Depending on the position of the switch, the set value and or that value is transferred, that the source that normally supplies the variable or the signal, is delivered.

Blocking and enabling of *messages with process information* and periodical information

The copy operation

- in the input-side process images or
- from the output-side process images

of

- *Messages with Process Information*
- Periodical Information

can be blocked and enabled. This can take place with the following granularity:

- per message
- all messages
- per periodical information
- each peripheral element

Changing the execution status of the open-/closed-loop control function

- Perform cold start or warm start of the resource
- Start and stop controller
- Perform cold start or warm start of a task
- Task halt and continue
- Program halt and continue

Test means

The available test means are:

(a) Halting of the execution due to a trigger condition ("Breakpoint")
(b) Execution of a task in cycle steps
(c) Controlling of the recording of the cyclic archive "Real Time Archive"

For each of the functions (a) and (c) a **Trigger Condition** is defined in the tool "CAEx *plus* Online Test". A trigger conditions consists of up to conditions. The conditions of a trigger condition are combined equal-ranking with AND or OR.

A condition compares a variable with a constant value to be specified:

Variable of the Type	Condition	<operator>			
BOOL	*variable* <operator> *value*	=		<>	
INT or REAL	*variable* <operator> *value*	<	=	<>	>

The trigger condition is assigned to either a task or the resource (= all tasks), depending on what one wishes to achieve:

- The function **Halt the execution due to a trigger condition** halts the task or resource, if the trigger condition is satisfied
- The function **Real Time Archive** switches over from Recording the Pre-History to Recording the Post-History, if the trigger condition is satisfied

The trigger condition is evaluated at the end of that task, to which it has been assigned; or at the end of every task, if it has been assigned to the resource.

Real Time Archive

The real time archive of the open-/closed-loop control function records variables (and signals) after every cycle, in order to make then available for display with the oscilloscope function of the TOOLBOX II tool "CAEx *plus* Online Test". Which variables are to be recorded is defined in the "CAEx *plus* Online Test". The recording can be controlled in such a way, that pre- and post-history are available for a post-mortem analysis.

For the recording of the variables there is a memory of 100,000 bytes available for each resource.

The recording can be terminated with:

- an operator input in "CAEX *plus* Online Test"

  The recording is terminated, the entire memory is available as pre-history.

- a definable Trigger Condition

  The real time archive switches from Recording the Pre-History to Recording the Post-History and continues to record until the memory is full. The division of the memory into pre- and post-history can be defined.

Which variables are to be recorded and at which periodicity, is determined in the "CAEX *plus* Online Test". The periodicity is determined by assigning the recording to a task. From its cycle time and the setting of how many cycles are to be omitted between the recordings, the recording times are produced and consequently the resolution of the display in the oscilloscope function.

$$Resolution_{[ms]} = cycle\ time_{[ms]}\ of\ the\ selected\ task * (number\ of\ cycles\ to\ be\ omitted + 1)$$

The time period for the pre- and post-history is dependent on the number of variables to be recorded (*# of Var*) and the aforementioned *resolution*:

$$Time\ period_{[ms]} = (100000 * resolution_{[ms]}) / ((\#\ of\ Var\ BOOL) + 2*(\#\ of\ Var\ INT) + 4*(\#\ of\ Var\ REAL))$$

Display Status Information

For each task the following informations are made available to the TOOLBOX II on request:

- the parameterized cycle time
- the current run time (in 10µs)
- the maximum run time (in 10µs)
- the number of time-outs that the system has registered

During the course of the interrogation, the current run time and the number of time-outs can be optionally reset.

4.3. System Services

System Services is a function package, that provides general functions and basic services in an automation unit, that other function packages require:

- Communication with the Engineering System
- Data Flow Control
- Addressing
- Time Management
- General Interrogation
- Self-Test
- Failure
- Diagnostic and Signaling
- Autonomy

4.3.1. Communication with the Engineering System (TOOLBOX II)

For the communication between the TOOLBOX II and the automation unit, in two respects there are different variants:

- physical connection of the TOOLBOX II with an automation unit
  - direct local via the serial Toolbox interface (TB) on the master control element
  - remote
    - serial via modems
    - Ethernet (TCP/IP, TELNET) and Terminal Server (serial)
    - Ethernet (TCP/IP, TELNET)

- logical connection of the TOOLBOX II with that automation unit, that is the subject of the engineering task:
  - local automation unit
    (that is that one, to which the physical connection exists, regardless in which of the above mentioned forms)
  - remote automation unit
    (automation unit that can be reached via the local automation unit; consistent remote communication according to IEC 60870-5-101 or -104 is required)

With the exception of the very first initialization procedures, all tasks are possible in each of the above mentioned variants, including for example:

- Parameter setting
- Diagnostic
- Test (for example online test of an application program of the open-/closed-loop control function)
- Load firmware, load parameters

4.3.2. Data Flow Control

The data flow control is that system function which co-ordinates the communication of messages within the automation unit.

This function supports:

- *Messages with Process Information*
- *Messages with System Information*

For the tracking of messages within an automation unit the following test functions are available:

- Data Flow Test
- Message Simulation

4.3.2.1. Messages with Process Information

IEC 60870-5 distinguishes between the following classes of messages. The type identification of each message provides information about the class to which a message belongs and with which methods it is to be distributed:

- *Messages with process information in monitor direction*
  - binary information, measured values, integrated totals and bit patterns
- *Messages with process information in control direction*
  - commands, setpoint values and bit patterns

The distribution of *messages with process information* takes place by way of routing (telecontrol) or assignment (open- / closed-loop control function) based on the message address and type identification in the message.

Messages with process information, that are to be transmitted to other automation units via protocol elements, are distributed with the help of the function Automatic Data Flow Routing.

For *messages with process information* that are to reach sinks within the automation unit - such as e.g. peripheral elements, open-/closed-loop control function - the routing information or assignments are automatically derived from parameters from OPM-inputs (datapoint address).

Predominantly used are message formats according to IEC 60870-5-101 / 104 in the public range with the exception of user data containers. Therefore, for their part the messages are compatible and interoperable with many other manufacturers.

Within the SICAM 1703 family, when using standard protocols the messages are compatible with the system families **Ax 1703** (AK 1703, AM 1703, AMC 1703, BC 1703) and **ACP 1703** (AK 1703 ACP, TM 1703 ACP, BC 1703 ACP).

Messages with process information have a 5-stage message address. Message addresses must be parameterized at the sources, such as e.g. peripheral elements, open-/closed-loop control function.

4.3.2.2. Addressing of the Process Information

Each data point is addressed in the automation network according to IEC 60870-5-101/104 by means of:

CASDU 1	Common address of the ASDU, octet 1
CASDU 2	Common address of the ASDU, octet 2
IOA 1	Information object address, octet 1
IOA 2	Information object address, octet 2
IOA 3	Information object address, octet 3

Messages with process information in monitor direction are source-addressed, *messages with process information in control direction* destination-addressed.

4.3.3. Addressing of Automation Units and System Elements

4.3.3.1. Addressing of Automation Units

An automation unit is addressed by means of:

- region number (0 .. 249) and
- component number (0 .. 255).

Within a system-technical plant each automation unit must be unambiguously addressed. Therefore a system-technical plant may consist of up to 64.000 automation units.

4.3.3.2. Addressing of System Elements

Within an automation unit, system elements are addressed by means of numbers for:

- Basic system elements
- Peripheral elements
- Protocol elements

4.3.4. Time Management

It is an integral element of the time management, that each automation unit and each system element, that has a time-dependent function to fulfill, can manage a clock with corresponding accuracy and resolution. Each automation unit has a central clock, the so-called time server.

4.3.4.1. Clock

In error-free operation the clocks are set once with the *time setting* operation and then run completely autonomous without any further time setting mechanisms.

Afterwards the *time synchronization* ensures, that all time servers in all automation units run synchronously.

All clocks within an automation unit are operated and synchronized by a central 10 ms clock pulse, that is generated by the time server of the automation unit with an accuracy of < 1 ms.

With a restart, all clocks begin to run unsynchronized with the value 0 hours. In other words, until the first *time setting* they have only a relative time, that is flagged with "invalid".

4.3.4.2. Time Setting and Time Synchronization

There are many options available for time setting and time synchronization:

- Direct serial connection of a DCF77 or a GPS time signal receiver
- Time setting over the communication (serial, LAN) from a master station,
  time synchronization by means of minute pulse of the GPS- or DCF77-receiver
- Time setting and time synchronization over a serial communication
- Time setting and time synchronization over LAN (Ethernet TCP/IP-NTP)

BC 1703 ACP itself can also provide further automation units with time over communication lines, and with serial standard protocols, can perform the time synchronization.

4.3.4.3. Automatic Time Tag (Time Stamp)

At every point in the system where messages with process information are generated, these messages can be provided with a time tag. Resolution and accuracy of the time tag are dependent on the function, that generates the time tag, and the system element in which this takes place.

The transfer of the data with standard protocols takes place with 7 octet time (in other words including date, with 1ms resolution) and priority-controlled.

4.3.5. General Interrogation

On startup and after faults in the system (communication faults, FIFO overflows), the participating automation units ensure, that the operation is resumed automatically in a coordinated manner.

This means, that the internal and external communication connections are set up (again). Under consideration of a multi-hierarchical network, all affected data and relevant system information are transmitted throughout the system from their sources though to their sinks for the purpose of updating the process images. This takes place with the initiation of a general interrogation to the corresponding part of the automation network, in which the error occurred.

4.3.6. Self-Test

The self-test is used for the protection against inadmissible operating states. Through a series of monitoring operations, defects of the hardware used or faulty behavior of the firmware are detected.

Depending on which test is concerned, a test is performed:

- during startup and/or
- continuously during operation

4.3.6.1. Monitoring of Hardware and Firmware

Monitoring	detects	Note
Watchdog monitoring	• Defect of the CPU used • Faulty behavior of the firmware	
IDLE monitoring	• Faulty behavior of the firmware or application program with endless loops • Excessively long firmware run times etc	
Code memory monitoring	• Defect of the storage medium used (Flash-PROM or Flash Card) • Undetected transmission errors when loading the program code	• Checksum
Parameter memory monitoring	• Defect of the storage medium used (Flash-PROM or Flash Card) • Undetected transmission errors when loading the parameters	• Checksum
Firmware self-monitoring	• Incorrect call parameters with system services and programming errors	
Shadow RAM	• Defect of the storage medium used (DRAM) • Defect of the DRAM refresh logic implemented	
RAM test with addressing errors check	• Defect of the storage medium used (DRAM) • Defect of the READ/WRITE equipment • Defect of the RAM´s in a defined area • Defect of the DRAM refresh logic implemented • Short or interruptions on the data and address bus	
Monitoring forbidden memory access	• Firmware errors	• CPU exception handling
Monitoring forbidden I/O access	• Firmware errors when accessing the I/O address range • Hardware errors	• CPU exception handling
Illegal Opcode	• Firmware errors with e.g. jump operations • Defect of the storage medium used (Flash-PROM) • Defect of the READ equipment • Short or interruptions on the data and address bus	• CPU exception handling
Stack Overflow	• Firmware errors	• CPU exception handling • Firmware

4.3.6.2. Monitoring the Data Integrity

Monitoring	detects	Note
Messages with spontaneous information objects on internal interfaces	• Defect of the storage medium used (FIFO) • Internal communication errors	• Checksum
Messages with periodical information on the Ax 1703 peripheral bus	• Communication errors on the Ax 1703 peripheral bus	• Hamming distance 4 • Horzontal and vertical parity

4.3.7. Failure

The system concept of failure management realized in SICAM 1703 ensures the individual marking of the data of failed parts of the system and the correct system and process behavior in the event of a fault.

For this the failure management provides:

- a system function for the failure detection (e.g. for modules/system elements, communication);
- derived from this the marking of the data points affected by the failure in the spontaneous communication with
  - other automation units
  - the open-/closed-loop control function
  - the peripheral elements
- periodical information that inform the open-/closed-loop control function which peripheral elements - and thereby which periodical information - are affected by the failure
- a parameter-settable behavior of peripheral elements with output function.

Consequently, for all data sinks (peripheral outputs, open-/closed-loop control function, control system) the state is available for every process information and – depending on requirements and functionality – corresponding measures can be initiated.

4.3.8. Diagnostic and Signaling

The diagnostic function manages the system states and error information detected by the individual functions and their monitoring operations.

It enables the display of process states, the internal system and fault information on the front panel of the modules and the local or remote diagnostic by means of the TOOLBOX II.

Each system element delivers its detected system- and error states to the master control element with supplementary information (e.g. cause of error, originator description). There they are saved in tables as current and stored information. These information items can be read out and displayed in detail locally or remotely with the help of the TOOLBOX II. The stored information can be acknowledged and can thus be updated again. For the purpose of better clarity, these tables are divided into classes:

A sum information about the detailed errors is transmitted from each BC 1703 ACP via the communication to the next automation units and is additionally managed there.

One BC 1703 ACP has up to two relay outputs available for signaling, each with normally open and normally closed contact.

Important detail or sum information is displayed by means of LEDs on the front panels of the system elements.

4.3.9. Autonomy

Autonomy means, that an autonomous basic system element and its supplementary system elements (protocol and peripheral elements) continue to function unaffected during the failure of the master control unit. This behavior can be set for each basic system element by means of a parameter.

On failure of the master control unit, data points are flagged with "not topical", which

- are acquired by other system elements in the automation unit and not over the particular peripheral or protocol elements;
- are acquired by other automation units that are not connected over the particular protocol elements.

After startup of the master control unit, the autonomous basic system element is synchronized without interruption to operation. Due to a general interrogation, the data points flagged with “not topical” on failure of the master control unit are updated.

5. Compatibility

Contents

5.1. Compatibility with the existing SICAM 1703 Product Family

Existing plants with Ax 1703 automation units (AK 1703 (Ax-Mode), AM 1703 (Ax-Mode), AMC 1703 (Ax-Mode)) can be networked without problems with automation units described in this document, either over the serial standard communication according to IEC 69870-5-101 or via Ethernet TCP/IP according to IEC 69870-5-104.

Networking of automation units described in this document with other automation units of the ACP 1703 platforms is a standard feature and therefore supported.

6. Technical Specification of the System

Contents

6.1. General Data

6.1.1. Mechanical Design

Basic Mounting Rack - 9 Slots	
Mechanical Design	Mounting rack for **rear panel installation**
Dimensions (rear panel installation)	Height approx. 291mm 6HE Width approx. 280mm 19" Depth approx. 258mm (power supply not installed) approx. 285 mm (power supply installed)
TB connector for TOOLBOX II	D-SUB 9pin, female (DIN 41652)
Connector for SI0 .. SI1 via CM-2837	RJ45 connector 8pin
Connector for SI0 .. SI3 via CM-2838	RJ45 connector 8pin
Type of protection according to IEC 60529	front, back IP 30 above IP 20
Weight	Approx. 5.8 kg no additional parts installed, however including 1 connection cable and 1 power supply

Basic Mounting Rack - 17 Slots	
Mechanical Design	Mounting rack for **19 inch and rear panel installation**
Dimensions (19" assembly)	Height approx. 291mm 6HE Width approx. 483mm 19" Depth approx. 258mm (power supply not installed) approx. 285 mm (power supply installed)
Dimensions (rear panel installation)	Height approx. 291mm 6HE Width approx. 483mm 19" Depth approx. 258mm (power supply not installed) approx. 285 mm (power supply installed)
TB connector for TOOLBOX II	D-SUB 9pin, female (DIN 41652)
Connector for SI0 .. SI1 via CM-2837	RJ45 connector 8pin
Connector for SI0 .. SI3 via CM-2838	RJ45 connector 8pin
Type of protection according to IEC 60529	• in case of rear panel installation with Wall fastening kit for CM-2835 AK 1703 ACP – front, back IP 30 – above IP 20 • in case of 19 inch (swing) frame installation with IP 20-Backside Cover kit for CM-2835 and FRBL19"-3HE Front Panel -RAL7032 – front, back IP 30 – above IP 20
Weight	Approx. 6.2 kg no additional parts installed, however including 1 connection cable and 1 power supply

Expansion Mounting Rack - 16 Slots			
Mechanical Design	Mounting rack for **19 inch and rear panel installation**		
Dimensions (19" assembly)	Height Width Depth	approx. 291mm approx. 483mm approx. 258mm approx. 285 mm	6HE 19" (power supply not installed) (power supply installed)
Dimensions (rear panel installation)	Height Width Depth	Approx. 291mm approx. 483mm approx. 258mm approx. 285 mm	6HE 19" (power supply not installed) (power supply installed)
Weight	Approx. 6.2 kg		no additional parts installed, however including 1 connection cable and 1 power supply

All Mounting Racks	
Peripheral connectors	Prefabricated peripheral cables CM-2890
Power supply connectors	Screw terminals for direct conductor assembly up to 2.5 mm^2 cross-section

6.1.2. Power Supply

The mounting racks CM-2832, CM-2835 and CM-2833 are to be equipped with 80W power supplies of the following types:

	Designation
PS-5620	Power Supply 24-60 VDC
PS-5622	Power Supply 110-220VDC, 230VAC

Information on the power supplies can be found in the PS-562x data sheet (see "Literature").

6.2. Environmental Conditions

6.2.1. Permitted Mechanical Stress

Parameter	Range	Testing Standard		Product Standard(s)	
Harmonic, sinusoidal					
Amplitude of the excursion 1 .. 9Hz Acceleration 9 .. 200Hz Acceleration 200 .. 500Hz	±0.035mm 10m/s^2 15m/s^2	IEC60068-2-6		IEC60870-2-2	Bm
Amplitude 10 .. 60Hz Acceleration 60 .. 150Hz	±0.035mm 0.5g	IEC60068-2-6		IEC60255-21-1	1
Shock					
Acceleration; 11ms duration	100m/s^2	IEC60068-2-27		IEC60870-2-2	Bm
Acceleration; 11ms duration	5g	IEC60068-2-27		IEC60255-21-2	1
Permanent shock					
Acceleration; 16ms duration	10g	IEC60068-2-29	1	IEC60255-21-2	1
Seismic harmonic, sinusoidal					
±3.5mm amplitude (horizontal) ±1.5mm amplitude (vertical) 1g acceleration (horizontal) 0.5g acceleration (vertical)	1 .. 8Hz 1 .. 8Hz 8 .. 35Hz 8 .. 35Hz	IEC60068-3-3		IEC60255-21-2	1

Transport

- The permitted mechanical stresses during transport depend on the transport packaging.
- The device packaging is not a transport packaging.

6.2.2. Permitted Climatic Environmental Conditions

This class is intended for indoor locations with temperature control and a wide range of relative humidity. The humidity is not controlled.

The devices can be exposed to sun and heat. They can be exposed as well to air flow caused by draught in buildings, e.g. by open windows or influences of technical processes. Condensation water, precipitations, water and icing do not occur.

Bedewing is possible for a short time (e.g. during the course of maintenance tasks).

Heating and cooling is used to maintain the necessary conditions, especially in case of great differences between indoor and outdoor climate.

The conditions of this class normally occur in living and working areas, e.g. in production rooms for electronic and electrotechnical products, telecontrol rooms, storage rooms for valuable and sensible devices.

Parameter	Range	Testing Standard		Product Standard(s)	
Minimum air temperature	-5°C	IEC 60068-2-1	Ad	IEC60870-2-2	C1
				IEC60654-1	C1
Maximum air temperature	+55°C	IEC 60068-2-2	Bd	IEC60870-2-2	C2
				IEC60654-1	C2
Temperature gradient	≤ 30°C/h			IEC60870-2-2	C2
				IEC60654-1	C2
Relative air humidity	5 .. 95%			IEC60870-2-2	C1
				IEC60654-1	C1
Absolute air humidity	≤ 29g/m³			IEC60870-2-2	C2
				IEC60654-1	C2
Air pressure	70 .. 106 kPa			IEC60870-2-2	C2
				IEC60654-1	C2
Storage and transport temperature	-30 ... +85°C				

6.2.3. Electrical Ambient Conditions

6.2.3.1. System Properties

Parameter	Value	Testing Standard		Product Standard(s)	
Immunity against discharge of static electricity (ESD)	8kV-L, 6kV-K	IEC61000-4-2	3	IEC60870-2-1	3
				IEC60255-22-2	
Immunity against electromagnetic fields	10V/m	IEC61000-4-3	3	IEC60870-2-1	3
				IEC60255-22-3	
Immunity against electromagnetic fields of GSM	10V/m	ENV50204	3	EN50082-2	-
				IEC61000-6-2	
Radio interference voltage - approx. peak vallue	79/73dBµV	CISPR22	A	IEC60870-2-1	A
				CISPR22	A
Radio interference voltage - mean value	66/60dBµV	CISPR22	A	IEC60870-2-1	A
				CISPR22	A
Radio interference field strength (10m)	40/47dBµV	CISPR22	A	IEC60870-2-1	A
				CISPR22	A
Immunity against electromagnetic fields 50Hz	100A/m	IEC61000-4-8	5	IEC60870-2-1	4
Immunity against induced HF voltage	10V	IEC61000-4-6	3	EN50082-2	-

The characteristics required by the European standards EN 50081-2 and IEC 61000-6-2 are covered by the values given.

6.2.3.2. Power Supplies

Parameters: Device Input		Value	Testing Standard		Product Standard(s)	
Voltage tolerance DC [1]		+30/-25%	IEC60870-2-1		IEC60870-2-1 IEC60654-2	>DC3 DC4
Voltage ripple DC [1]		≤ 15%	IEC60870-2-1		IEC60870-2-1 IEC60255-11	>VR3
Voltage tolerance AC [2]		+10/-15%	IEC60870-2-1		IEC60870-2-1 IEC60654-2	AC2 AC3
Frequency tolerance AC		±5%	IEC60870-2-1		IEC60870-2-1 IEC60654-2	F3
Harmonic content		< 20%	IEC60870-2-1		IEC60870-2-1 IEC60654-2	>H2
Harmonic current			IEC61000-3-2	D	IEC60870-2-1	A=B
Interruption time		≤50ms	IEC61000-4-11		IEC60870-2-1 IEC60255-11	>1
Dielectric test up to 60V nominal voltage		2.5kVrms	IEC60255-5		IEC60870-2-1 IEC60255-5	VW3
Dielectric test > 60V nominal voltage against SELV-circuits		3kVeff	IEC60255-5		IEC60950 IEC60950	
Impulse voltage 1.2/50µs	common	5kVs	IEC60255-5		IEC60870-2-1 IEC60255-5	VW3
Impulse voltage 1.2/50µs	normal	5kVs	IEC60255-5		IEC60870-2-1 IEC60255-5	VW3
HF test	common	2.5kVs	IEC61000-4-12	3	IEC60870-2-1 IEC60255-22-1	3-4
HF test	normal	2.5kVs	IEC61000-4-12	>3	IEC60870-2-1 IEC60255-22-1	>3-4
Fast transient burst	common	4kVs	IEC61000-4-4	4	IEC60870-2-1 IEC60255-22-4	4 A
1.2/50µs surge (DC type)	common	4kVs	IEC61000-4-5	4	IEC60870-2-1 IEC60255-22-5	4
1.2/50µs surge (DC type)	normal	4kVs	IEC61000-4-5	4	IEC60870-2-1 IEC60255-22-5	>4
1.2/50µs surge (AC type)	common	4kVs	IEC61000-4-5	4	IEC60870-2-1 IEC60255-22-5	4
1.2/50µs surge (AC type)	normal	4kVs	IEC61000-4-5	4	IEC60870-2-1 IEC60255-22-5	>4
10/700µs surge	common	2.5kVs	IEC61000-4-5	3	IEC60870-2-1	3-4
100/1300µs surge	normal	1,3xUn	EN50178		IEC60870-2-1	-
Starting current		S1	IEC60870-4		IEC60870-4	S1

1) referring to supply voltage rated values: 24VDC / 48VDC / 60VDC, 110VDC / 220VDC

2) referring to supply voltage rated values: 115V_{rms} AC / 230V_{rms} AC

6.2.3.3. Digital and Analog Standard I/O´s

Parameter		value	Testing Standard		Product Standard(s)	
Dielectric test		1.5kVrms	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Impulse voltage 1.2/50µs	common	2.5kVs	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Impulse voltage 1.2/50µs	normal	2.5kVs	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
HF test	common	1kVs	IEC61000-4-12		IEC60870-2-1	2
					IEC60255-22-1	
HF test	normal	1kVs	IEC61000-4-12		IEC60870-2-1	>2
					IEC60255-22-1	
Fast transient burst	common	2,0kVs	IEC61000-4-4	3	IEC60870-2-1	3
					IEC60255-22-4	B
1.2/50µs surge	common	2,0kVs	IEC61000-4-5	3	IEC60870-2-1	3
					IEC60255-22-5	
1.2/50µs surge	normal	2,0kVs	IEC61000-4-5	3	IEC60870-2-1	4
					IEC60255-22-5	

Deviating values for DO-2210

Parameter		Value	Testing Standard		Product Standard(s)	
HF test	common	2.5kVs	IEC61000-4-12	3	IEC60870-2-1	3-4
					IEC60255-22-1	
Fast transient burst	common	4,0kVs	IEC61000-4-4	4	IEC60870-2-1	4
					IEC60255-22-4	A

Deviating values for DI-2110 and DI-2111

Parameter		Value	Testing Standard		Product Standard(s)	
Dielectric test up to 60V nominal voltage		2.5kVrms	IEC60255-5		IEC60870-2-1	VW3
					IEC60255-5	
Dielectric test > 60V nominal voltage against SELV-circuits		3kVrms	IEC60255-5		IEC60950	
					IEC60950	
Impulse voltage 1.2/50µs	common	5kVs	IEC60255-5		IEC60870-2-1	VW3
					IEC60255-5	

6.2.3.4. Serial Communication V.24 / V.28

Parameter		Value	Testing Standard		Product Standard(s)	
Dielectric test		1.5kVrms	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Impulse voltage	common	2.5kVs	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Fast transient burst	common	2kVs	IEC61000-4-4	3	IEC60870-2-1	3
					IEC60255-22-4	B

Valid for a distance ≤ 30m, screened

6.2.3.5. Serial Communication Profibus DP

Parameter		Value	Testing Standard		Product Standard(s)	
Dielectric test		1.5kVrms	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Impulse voltage	common	2.5kVs	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Fast transient burst	common	2kVs	IEC61000-4-4	3	IEC60870-2-1	3
					IEC60255-22-4	B

6.2.3.6. LAN Communication 10/100Base-T

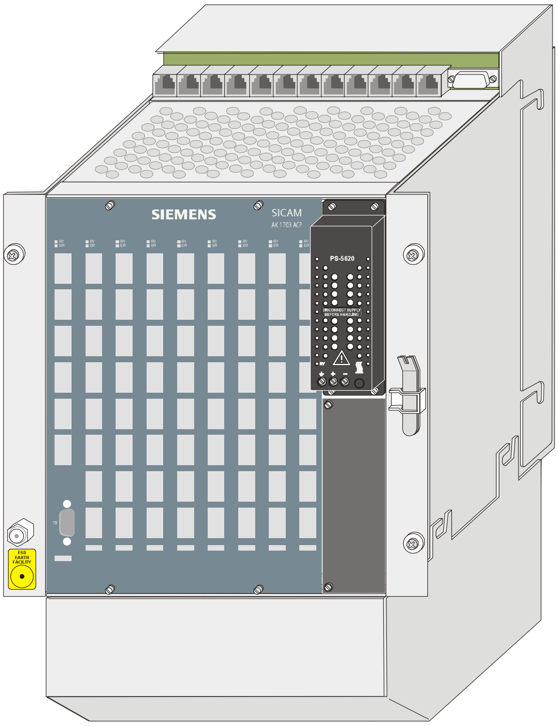
Parameter		Value	Testing Standard		Product Standard(s)	
Dielectric test		1.5kVrms	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Impulse voltage	common	2.5kVs	IEC60255-5		IEC60870-2-1	>VW2
					IEC60255-5	
Fast transient burst	common	2kVs	IEC61000-4-4	3	IEC60870-2-1	3
					IEC60255-22-4	B

Valid for a distance ≤ 100m, screened

7. Mechanical Design

Contents

7.1. CM-2832 Basic Mounting Rack, 9 Slots

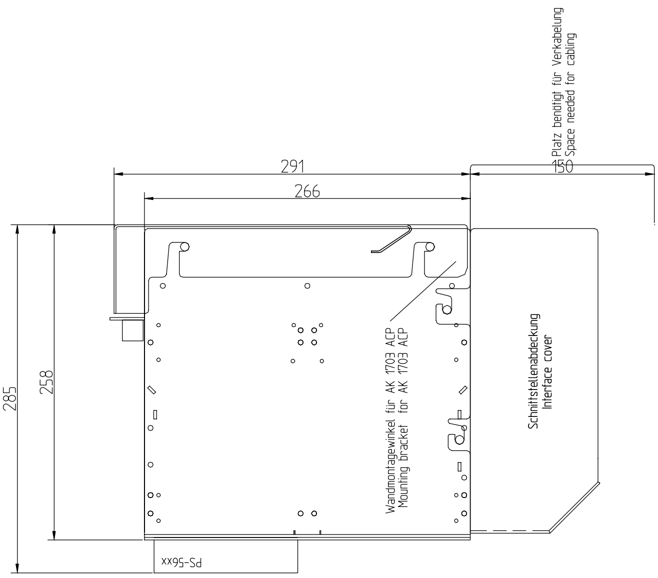


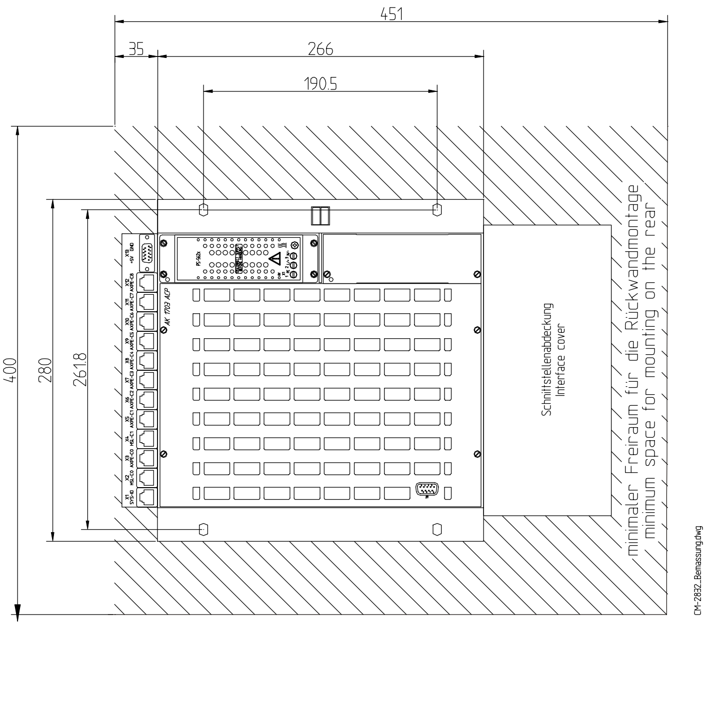
The picture above shows a mounting rack for rear panel installation.

In the following you can find information on dimensions and space requirement.

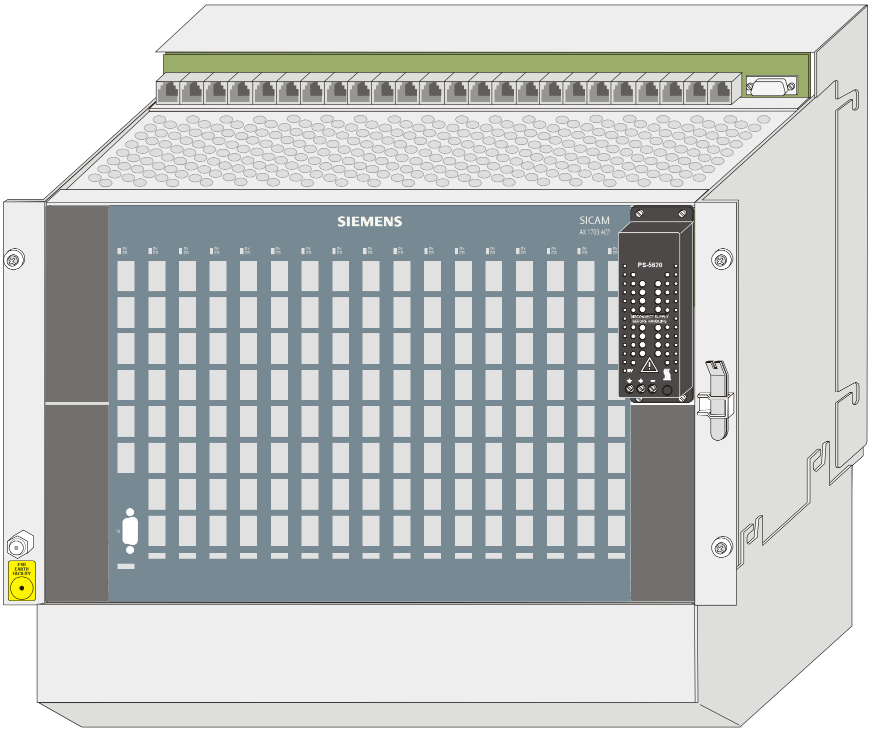
Further information can be found in the relevant data sheet, see Literature.

7.1.1. Dimensions and Space Requirements - Rear Panel Installation





7.2. CM-2835 Basic Mounting Rack, 17 Slots



As an example, the picture above shows a mounting rack, equipped for rear panel installation.

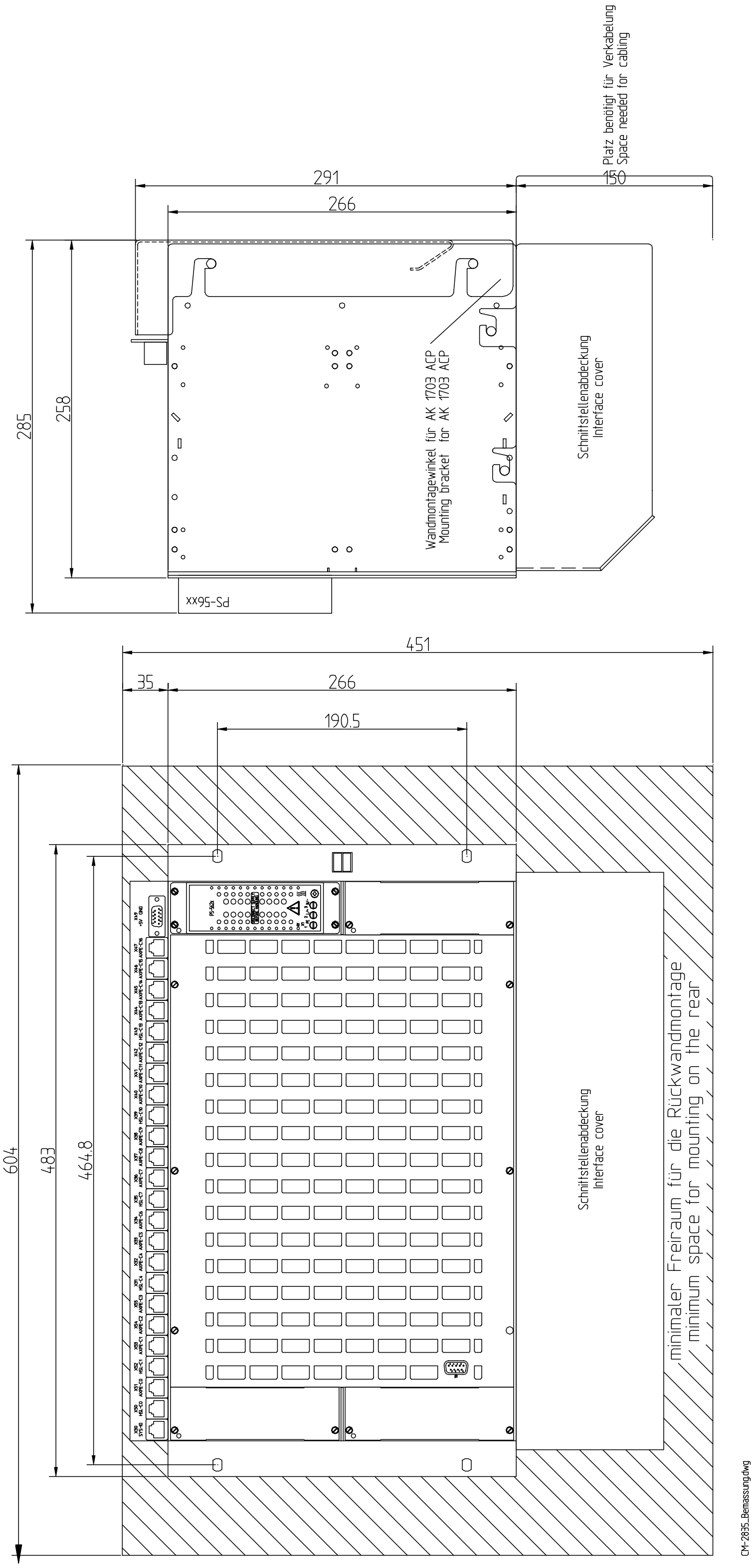
In the following you can find information on dimensions and space requirement.

Further information can be found in the relevant data sheet, see Literature.

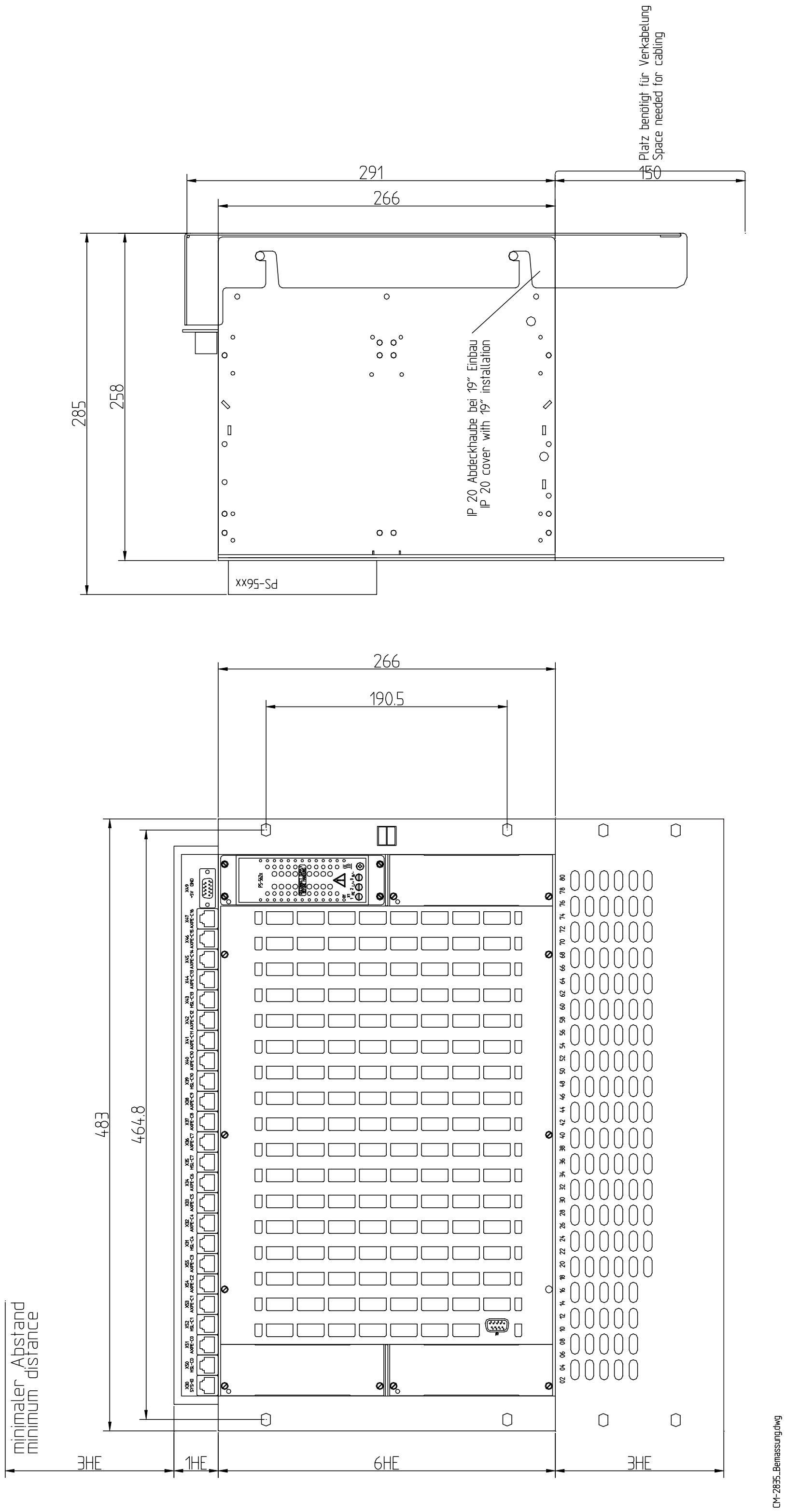
7.2.1. Accessories

Wall fastening kit for CM-2835 AK 1703 ACP	(for rear panel installation)	TC2-008
IP 20-Backside Cover kit for CM-2835	(for 19 inch (swing) frame installation)	TC2-050
FRBL19"-3HE Front Panel -RAL7032	(for 19 inch (swing) frame installation)	MC2-003-1

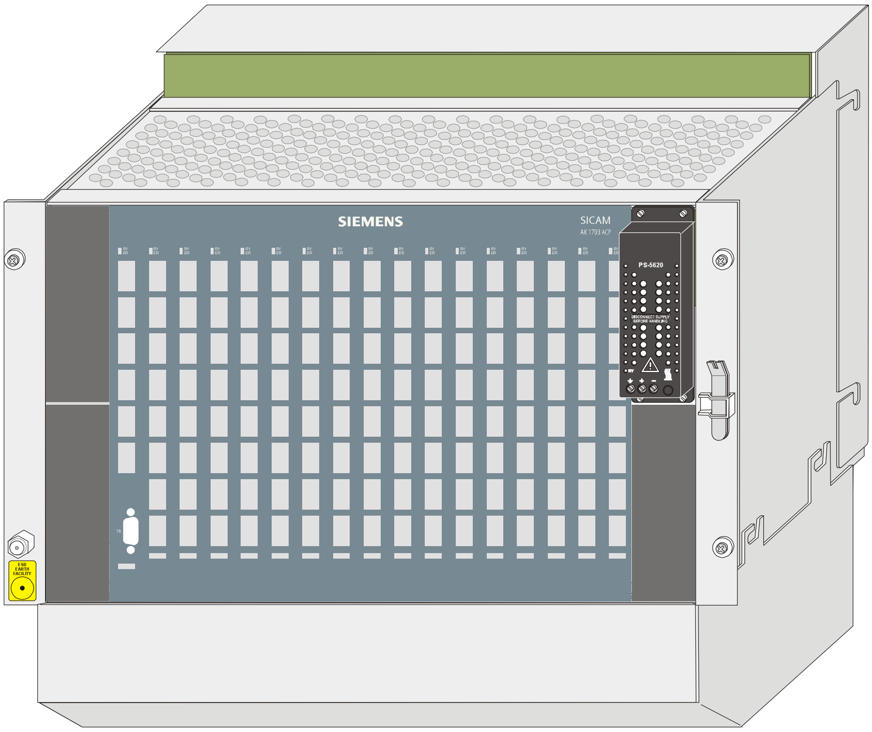
7.2.2. Dimensions and Space Requirements - Rear Panel Installation



7.2.3. Dimensions and Space Requirements - 19 inch (swing) frame installation



7.3. CM-2833 Expansion Mounting Rack, 16 Slots



As an example, the picture above shows a mounting rack, equipped for rear panel installation.

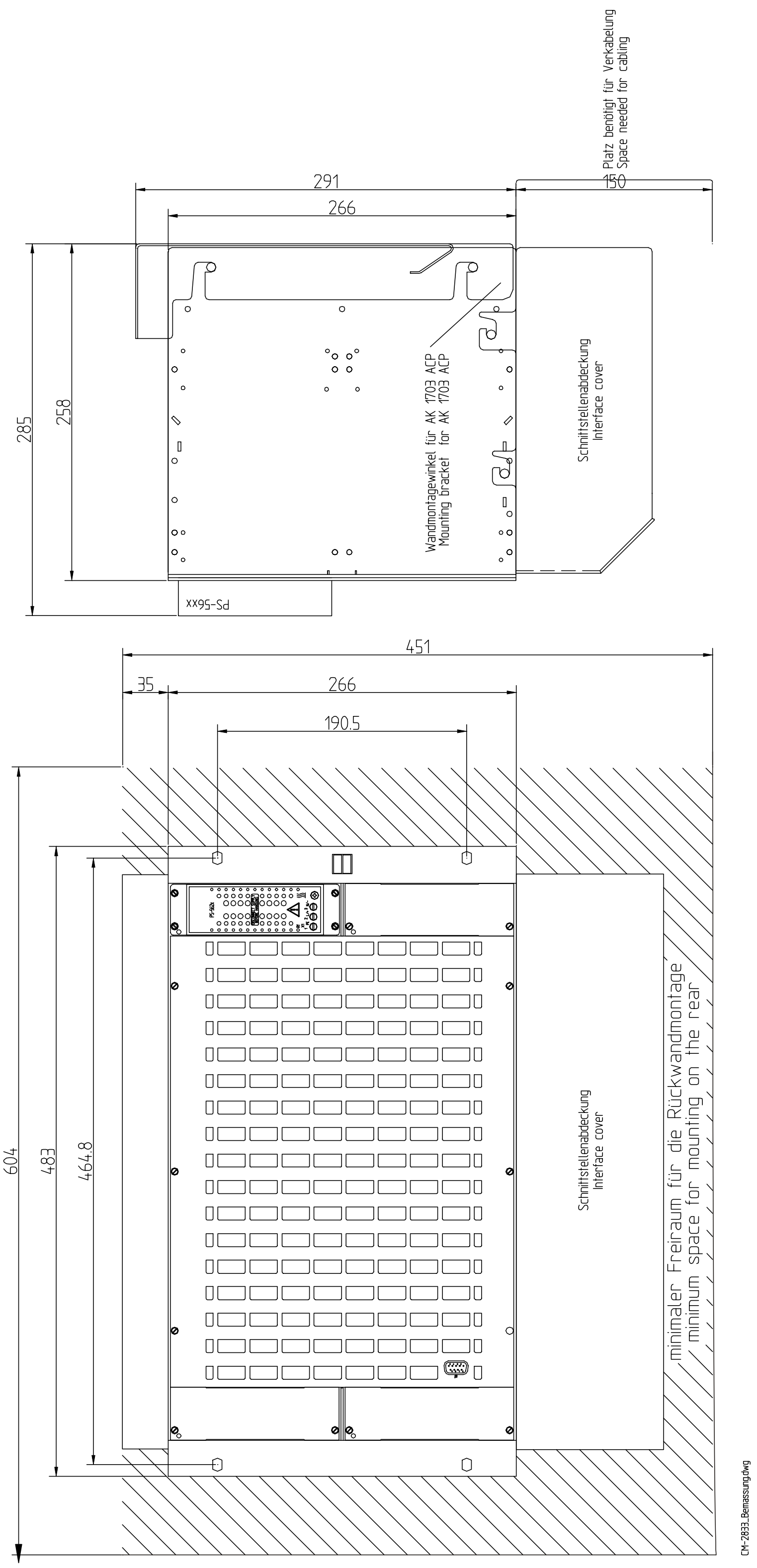
In the following you can find information on dimensions and space requirement.

Further information can be found in the relevant data sheet, see Literature.

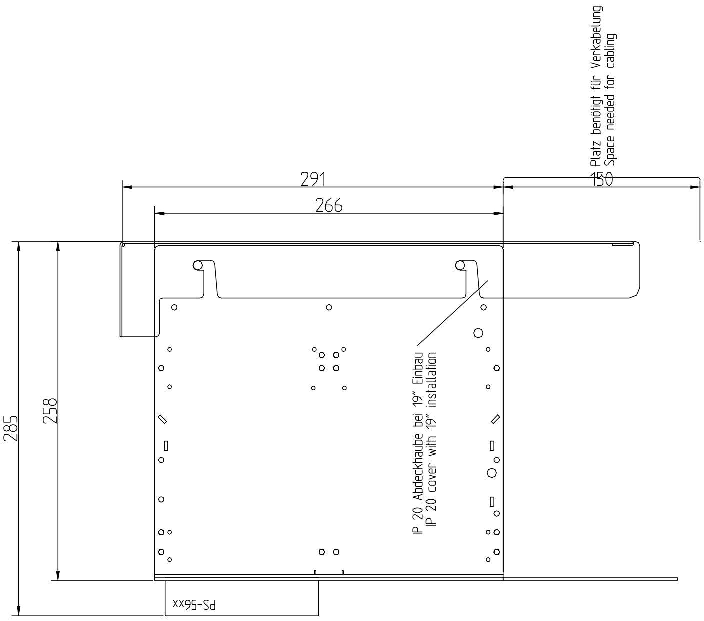
7.3.1. Accessories

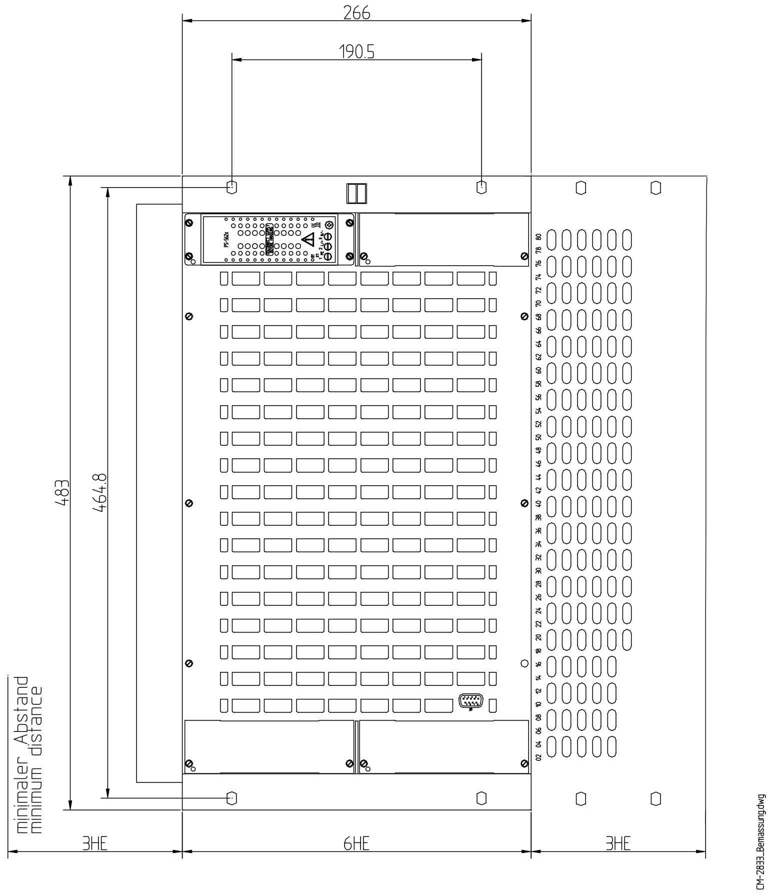
Wall fastening kit for CM-2835 AK 1703 ACP	(for rear panel installation)	TC2-008
IP 20-Backside Cover kit for CM-2835	(for 19 inch (swing) frame installation)	TC2-050
FRBL19"-3HE Front Panel -RAL7032	(for 19 inch (swing) frame installation)	TF3-018

7.3.2. Dimensions and Space Requirements - Rear Panel Installation



7.3.3. Dimensions and Space Requirements - 19 inch (swing) frame installation





8. Ordering Information

Contents

8.1. Mounting Racks and Connection Technique

	Description
CM-2832	AK 1703 ACP Mounting Rack with 9 Slots
	AK 1703 ACP mounting rack (44WU, 8HU) with room for • 9 double-Euroformat modules, thereof — 1 master control element and — up to 8 processing, communication, and peripheral elements • up to 2 power supplies PS-562x Connectors: • 9x Ax 1703 peripheral bus • 1x for redundancy switch SCA 1703-RED • 2x redundancy link The rack can be used for rear panel installation.

	Description
CM-2835	AK 1703 ACP Mounting Rack with 17 Slots
	AK 1703 ACP mounting rack (84 WU, 8 HU) with room for • 17 double-Euroformat modules, thereof — 1 Master control element and — up to 16 processing, communication, and peripheral elements • up to 4 power supplies PS-562x Connectors: • 17x Ax 1703 peripheral bus • 1x for redundancy switch SCA 1703-RED • 6x redundancy link The rack can be used for • rear panel or • 19 inch (swing) frame installation.

	Description
CM-2833	AK 1703 ACP Expansion Mounting Rack
	AK 1703 ACP expansion mounting rack (84 WU, 8 HU) with room for • 16 peripheral elements • up to 4 power supplies PS-562x Connectors: • 2x Ax 1703 peripheral bus The rack can be used for • rear panel or • 19 inch (swing) frame installation.

	Description
CM-2837	Connection board for CP-2010 (SI0-SI3, SYS)
	Connection PCB for CP-2010 for the connection of • the serial interfaces (SI0, SI1) or • the LAN/WAN interface (SI1) or • the field bus interface (SI0) via two RJ45 connectors. The pin assignment of the RJ45 connectors depends on which serial interface module (SIM) is installed. Additional connectors are provided for the two fault outputs (WD, ER) and the synchronisation input (SYNC).

	Description
CM-2838	Connection board for CP-2012 (SI0-SI3)
	Connection PCB for CP-2012 for the connection of • the serial interfaces (SI0, SI1, SI2, SI3) or • the LAN/WAN interfaces (SI1, SI3) or • the fieldbus interfaces (SI0, SI2) via four RJ45-connectors. The pin assignment of the RJ45-connectors depends on which serial interface modules (SIM) have been installed.

	Description
CM-2890	Peripheral Connection Kit
	Prefabricated connection kit for all peripheral elements, useable in AK 1703 ACP rack. • Configured as crimp connection with grip sleeve • 5m cable 50x2x0.5 not halogen-free • Connected on device side

	Description
WALL-FASTENING-KIT	Wall fastening kit for CM-2835 AK 1703 ACP
	Wall fastening kit for CM-2835 AK 1703 ACP. To build in the 19"-version onto a mounting plate. A swing frame is not nessesary.

	Description
IP20-COVER-KIT	IP 20-Backside Cover kit for CM-2835
	Back side cover for CM-2835. Material: Aluminium 1,5mm

	Description
FRBL19"-3HE-RAL7032	FRBL19"-3HE Front Panel -RAL7032
	Aluminium front panel 19"-inches 3HU Color: RAL 7032

8.2. Basic System Elements

	Description
CP-2010/CPC25	System Functions, Processing and Communication
	Master control element with 2 microprocessors and: • up to 2 communication interfaces via one installable SM-25xx serial interface modules: – serial (end-end, multi-point, dial-up traffic) – LAN/WAN (Ethernet) – Profibus-DP • communication with up to 16 peripheral elements via the serial Ax 1703 peripheral bus • hub functionality for up to 16 additional processing- and/or communication elements via the internal node bus • open-/closed-loop control functions, freely definable with CAEx plus, IEC 61131-3 compliant • local and remote engineering, diagnostic, and test using TOOLBOX II • data are kept on a Flash Card for Plug&Play module replacement • function and failure indication via LED

	Description
CP-2012/PCCE25 **CP-2017/PCCX25**	Processing & Communication
	Processing and communication element with: • up to 4 communication interfaces via installable SM-25xx serial interface modules: – serial (end-end, multi-point, dial-up traffic) – LAN/WAN (Ethernet) – Profibus-DP • communication with up to 16 peripheral elements via the serial Ax 1703 peripheral bus • open-/closed-loop control functions, freely definable with CAEx plus, IEC 61131-3 compliant • redundancy with doubled CP-2012/PCCE25 supported • autonomy: function maintained even in case of master control element failure • local and remote engineering, diagnostic, and test using TOOLBOX II • function and failure indication via LED

8.3. Peripheral Elements

	Description
DI-2100/BISI25	Binary signal input (8x8, 24-60VDC)
	System element for acquisition and processing according to IEC 60870-5-101/104, of • up to 64 single-point information units, or • up to 32 double-point information units, or • up to 64 integrated totals via count pulses, or • a combination thereof with following features: • 64 inputs (8 groups) • galvanically insulated by optocouplers • each group has a common return • signal voltage 24-60VDC • filter-equipped inputs circuits • for each group, one additional input for power monitoring • integrated totals not power-fail safe • indication of function and states of the inputs via LEDs

	Description
DI-2110/BISI26	Binary signal input (8x8, 24-60VDC)
	System element for acquisition and processing according to IEC 60870-5-101/104, of • up to 64 single-point information units, or • up to 32 double-point information units, or • up to 64 integrated totals via count pulses, or • a combination thereof with following features: • 64 inputs + 8 auxiliary inputs (8 groups) • galvanically insulated by optocouplers • each group has a common return • signal voltage 24-60VDC • settable polarity and thresholds for each group • filter-equipped inputs circuits • for each group, one additional input for power monitoring • indication of function and states of the inputs via LEDs • binary information is acquired with a 1ms real-time resolution

	Description
DI-2111/BISI26	Binary signal input (8x8, 110/220VDC)
	System element for acquisition and processing according to IEC 60870-5-101/104, of • up to 64 single-point information units, or • up to 32 double-point information units, or • up to 64 integrated totals via count pulses, or • a combination thereof with following features: • 64 inputs + 8 auxiliary inputs (8 groups) • galvanically insulated by optocouplers • each group has a common return • signal voltage 110/220VDC • parameter settable polarity and thresholds for each group • filter-equipped inputs circuits • for each group, one additional input for power monitoring • indication of function and states of the inputs via LEDs • binary information is acquired with a 1ms real-time resolution

	Description
DO-2201/BISO25	Binary Output (Transistor, 40x1, 24-60VDC)
	System element for processing and output according to IEC 60870-5-101/104, of • up to 40 single-point information units with following features: • 40 digital outputs • with respect to galvanical insulation, the outputs are partitioned into 8 groups, 2 outputs each, and 8 groups, 3 outputs each • galvanical insulation — between groups 1.5kVeff — within a group 80V • the potential that shall be switched (plus or minus) can be determined for each output by external circuitry • all outputs are overload-proof and proof against continued short-circuit • any 2 outputs can be connected in parallel to increase the switching capacity • if an output short-circuits, it does not affect on other outputs

	Description
AI-2300/PASI25	Analog in-/output (16x ±20mA + 4x2 opt. expans.)
	System element for • acquisition and processing of analog values and integrated totals • processing and output of analog values according to IEC 60870-5-101/104, with the following features: • 16 analog inputs, galvanically insulated from logic (±20mA) • optionally, via 1 to 4 input/output modules, galvanically insulated from logic, 2 to 8 — analog inputs (±20mA, Pt100) — analog outputs (±20mA) — pulse inputs (24-60VDC) • indication of process information via LEDs

	Description
DO-2210/PCCO2X	Checked command output 24-60 VDC
	System element for processing and output (checked command output) according to IEC 60870-5-101/104, of • up to 32 pulse commands (2-pole), or • up to 64 pulse commands (1-pole or 1½-pole), or • a combination thereof, with • internal checks (IC1) • resistance check (RC1) (optional, submodule SM-2506 required) The system element has the following features: • 64 relay outputs (2 groups), and — 2 group outputs — 2 pulse outputs • each group has a common return • switching voltage 24-60VDC • each group may have its own fused circuit • the pulse outputs are current-limited electronically

8.4. Protocol Elements

	Description
SM-2551/PROTOCOL	2 Protocol Elements (2x V.28)
	2 protocol elements based on a Serial Interface Module (SIM) with two serial interfaces: Protocol element (serial) • SIEMENS standard protocols according to IEC 60870-5-101/103 for — point-to-point traffic — multi-point traffic — dial-up traffic • protocols subject to license • with byte-asynchronous or byte-isochronous puls code modulation • all interface signals (RS-232) and all interfaces are galvanically insulated from each other • protocol selectable per interface The SIM can be plugged on master control and communication elements of ACP 1703 and Ax 1703 platforms

	Description
SM-2556/PROTOCOL	2 Protocol Elements (V.28 + Ethernet TCP/IP el.)
	2 protocol elements based on a Serial Interface Module (SIM) with one serial and one LAN interface: Protocol element (serial) • SIEMENS standard protocols according to IEC 60870-5-101/103 for — point-to-point traffic — multi-point traffic — dial-up traffic • Protocols subject to license • With byte-asynchronous or byte-isochronous pulse code modulation • All interface signals (RS232) and all interfaces are galvanically insulated from each other • Protocol selectable Protocol element (LAN) • Standard according to IEC 60870-5-104 — Ethernet / Fast Ethernet 10/100 MBit/s, IEEE 802.3, 10/100Base-TX, electrical — TCP/IP — data formats according to IEC 60870-5-104 — time synchronisation via network time protocol (NTP) The SIM can be attached to master control and communication elements of ACP 1703 platforms

	Description
SM-2546/PROTOCOL	2 Protocol Elements (V.28 + Ethernet TCP/IP opt.)
	2 protocol elements based on a Serial Interface Module (SIM) with one serial and one LAN interface: Protocol element (serial) • SIEMENS standard protocols according to IEC 60870-5-101/103 for — point-to-point traffic — multi-point traffic — dial-up traffic • Protocols subject to license • With byte-asynchronous or byte-isochronous pulse code modulation • All interface signals (RS232) and all interfaces are galvanically insulated from each other • Protocol selectable Protocol element (LAN) • Standard according to IEC 60870-5-104 — Fast Ethernet 100Mbit/s, IEEE 802.3, 100Base-Fx, optical — TCP/IP — data formats according to IEC 60870-5-104 — time synchronisation via network time protocol (NTP) The SIM can be attached to the master control element of a BC 1703 ACP

	Description
SM-2554/ET02	Protocol Element for Ethernet TCP/IP electrical
	Protocol element for communication via LAN • According to IEC 60870-5-104 — Ethernet / Fast Ethernet 10/100 Mbit/s, IEEE 802.3, 10/100Base-TX, electrical — TCP/IP — data formats according to IEC 60870-5-104 — time synchronisation via network time protocol (NTP) The protocol element can be attached to master control and communication elements of ACP 1703 platforms

	Description
SM-2545/DPM00	Protocol Element for Profibus DP
	Protocol element for Profibus DP communication • Profibus DP Master according to EN 50170-2 — RS-485 unbalanced, up to 12 Mbps — up to 100 Profibus DP slaves per master — 1000 data points in receive direction — 900 data points in transmit direction • the SM-2545/DPM00 implementation is certified by Profibus Nutzeror-ganisation The protocol element can be plugged on master control and communication elements of ACP platforms

Literature

Folder AK 1703 ACP	MC2-009-2
ACP 1703 Common Functions System and Basic System Elements	DC0-015-2
SICAM 1703 Common Functions Peripheral Elements according to IEC 60870-5-101/104	DC0-011-2
ACP 1703 Platforms Configuration Automation Units and Automation Networks	DC0-021-2
Data Sheet CP-2010/CPC25	MC2-001-2
Data Sheet CP-2012/PCCE25	MC2-003-2
Data Sheet DI-2100/BISI25	MC0-009-2
Data Sheet DI-211x/BISI26	MC0-015-2
Data Sheet DO-2210/PCCO2X	MC0-011-2
Data Sheet DO-2201/BISO26	MC0-025-2
Data Sheet AI-2300/PASI25	MC0-017-2
Data Sheet MX-2400/USIO2x	MC0-027-2
Data Sheet SM-2541/PROTOCOL	MC0-001-2
Data Sheet SM-2551/PROTOCOL	MC0-003-2
Data Sheet SM-25x4/ET02	MC0-005-2
Data Sheet CM-2832	MC2-005-2
Data Sheet CM-2833	MC2-014-2
Data Sheet CM-2835	MC2-007-2
Data Sheet PS-562x	MC0-013-2